

Geological Society of America Special Papers

Tectonic terranes of the Chortis block based on integration of regional aeromagnetic and geologic data

Robert D. Rogers, Paul Mann and Peter A. Emmet

Geological Society of America Special Papers 2007;428;65-88
doi: 10.1130/2007.2428(04)

Email alerting services

click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe

click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Special Papers

Permission request

click <http://www.geosociety.org/pubs/copyrt.htm#gsa> to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

The Geological Society of America
Special Paper 428
2007

Tectonic terranes of the Chortis block based on integration of regional aeromagnetic and geologic data

Robert D. Rogers*

Paul Mann

Institute for Geophysics, Jackson School of Geosciences, University of Texas at Austin, J.J. Pickle Research Campus, Bldg. 196 (ROC), 10100 Burnet Road (R2200), Austin, Texas 78758-4445, USA

Peter A. Emmet

Cy-Fair College, Fairbanks Center, 14955 Northwest Freeway, Houston, Texas 77904, USA

ABSTRACT

An aeromagnetic survey of Honduras and its northeastern Caribbean coastal area covering a continuous area of 137,400 km² was acquired by the Honduran government in 1985 and provided to the University of Texas at Austin for research purposes in 2002. We correlate regional and continuous aeromagnetic features with a compilation of geologic data to reveal the extent, structural grain, and inferred boundaries of tectonic terranes that compose the remote and understudied, Precambrian-Paleozoic continental Chortis block of Honduras. A regional geologic map and a compilation of isotopic age dates and lead isotope data are used in conjunction with and geo-referenced to the aeromagnetic map. These combined data provide a basis for subdividing the 531,370 km² Chortis block into three tectonic terranes with distinctive aeromagnetic expression, lithologies, structural styles, metamorphic grade, isotopically and paleontologically determined ages, and lead isotope values: (1) The Central Chortis terrane occupies an area of 110,600 km², exhibits a belt of roughly east-west-trending high magnetic values, and exposes small, discontinuous outcrops of Grenville to Paleozoic continental metamorphic rocks including greenschist to amphibolite grade phyllite, schist, gneiss, and orthogneiss that have been previously dated in the range of 1 Ga to 222 Ma; the northern 59,990 km² margin of the Central Chortis terrane along the northern Caribbean coast of Honduras exhibits an irregular pattern of east-west-trending magnetic highs and lows that correlates with an east-west-trending belt of early Paleozoic to Tertiary age metamorphic rocks intruded by Late Cretaceous and early Cenozoic plutons in the range of 93.3–28.9 Ma. (2) The Eastern Chortis terrane occupies an area of 185,560 km², exhibits belts of roughly northeast-trending high magnetic values, and correlates with outcrops of folded and thrustured Jurassic metasedimentary phyllites and schists forming a greenschist-grade basement; we propose that the Eastern and Central terranes are distinct terranes based on the strong differences in

*Present address: Department of Geology, California State University, Stanislaus, 801 W. Monte Vista Dr., Turlock, California 95832, USA; rrogers@geology.csustan.edu.

Rogers, R.D., Mann, P., and Emmet, P.A., 2007, Tectonic terranes of the Chortis block based on integration of regional aeromagnetic and geologic data, *in* Mann, P., ed., *Geologic and tectonic development of the Caribbean plate boundary in northern Central America*: Geological Society of America Special Paper 428, p. 65–88, doi: 10.1130/2007.2428(04). For permission to copy, contact editing@geosociety.org. ©2007 The Geological Society of America. All rights reserved.

their structural style and aeromagnetic grain, sedimentary thickness, metamorphic grade, and lead isotope values. (3) The Southern Chortis terrane occupies an area of 120,100 km², contains one known basement outcrop of metaigneous rock, exhibits a uniformly low magnetic intensity that contrasts with the rest of the Chortis block, and is associated with an extensive area of Miocene pyroclastic strata deposited adjacent to the late Cenozoic Central American volcanic arc. The outlines of the terranes as constrained by the aeromagnetic, lithologic, age, and lead isotope data are restored to their pre-early Eocene position along the southwestern coast of Mexico by a 40° clockwise rotation and 1100 km of documented post-early Eocene (ca. 43 Ma) left-lateral offset along the strike-slip faults of the northern Caribbean strike-slip plate boundary. The inner continental and outboard oceanic terranes of Chortis and the 120,100 km² Siuna terrane to the south trend roughly north-south and align with terranes of similar magnetic trend, lithology, age, and crustal character in southwestern Mexico. Additional progress in mapping and isotopic dating is needed for the proposed Chortis terranes in Honduras in order to constrain this proposed position against much better mapped and dated rocks in southwestern Mexico.

Keywords: tectonic terranes, Chortis block, Caribbean plate, tectonics, regional geology.

INTRODUCTION

Tectonic Significance

The Chortis block of northern Central America (Honduras, Nicaragua, El Salvador, southern Guatemala, and part of the Nicaragua Rise) forms the only emergent area of Precambrian to Paleozoic continental crust on the present-day Caribbean plate (DeMets *et al.*, this volume) and has long been recognized as an important constraint on the tectonic origin and Cretaceous-Cenozoic displacement history of the Caribbean plate (Gose and Swartz, 1977; Pindell and Dewey, 1982; Case *et al.*, 1984; Dengo, 1985; Case *et al.*, 1990). (Fig. 1A). Largely due to the paucity of information from the Chortis block, three different tectonic models have been proposed to explain the tectonic origin of the Chortis block and its geologic relationship to Precambrian to Paleozoic rocks of similar lithologies and metamorphic grades from southwestern Mexico (Campa and Coney, 1983; Sedlock *et al.*, 1993; Dickinson and Lawton, 2001; Solari *et al.*, 2003; Keppie, 2004; Keppie and Morán-Zenteno, 2005; Talavera-Mendoza *et al.*, 2005) and northern Guatemala (Ortega-Gutierrez *et al.*, 2004).

The paucity of geologic and age isotopic information from Honduras that encompasses most of the Chortis block, is limited to a few studies, including Fakundiny (1970), Horne *et al.* (1976a), Simonson (1977), Sundblad *et al.* (1991), and Manton (1996), and can be attributed to the remoteness and inaccessibility of many parts of the country to geologic research (Fig. 1B). Because of its mountainous terrain, sparse population (62 people/km²), and poor road network, only ~15% of the geology of Honduras has been systematically mapped by field geologists at a scale of 1:50,000 (Fig. 2). The remaining 85% of the country remains either completely unmapped or subject to reconnaissance mapping and spot

sampling usually confined to major roads (e.g., Mills *et al.*, 1967; Finch, 1981; Manton and Manton, 1984, 1999).

Previous Tectonic Models for the Origin of the Chortis Block

Mexico-Derived Model for Chortis

The first model to explain the tectonic history of the Chortis block invokes large-scale strike-slip motion on the Cayman-Motagua-Polochic strike-slip fault system that presently forms the northern edge of the Chortis block (Figs. 1A and 1B). In this model, the Chortis block is detached from its pre-middle Eocene position along the southwestern coast of Mexico and moved eastward by ~1100 km of left-lateral strike-slip motion and ~30–40° of large-scale, counterclockwise rotation (Gose and Swartz, 1977; Karig *et al.*, 1978; Pindell and Dewey, 1982; Gose, 1985; Rosencrantz *et al.*, 1988; Pindell and Barrett, 1990; Sedlock *et al.*, 1993; Dickinson and Lawton, 2001; Pindell *et al.*, 2006) (Figs. 1A and 1B). Previous onland and offshore mapping and isotopic age dating studies have supported the presence of a linear, strike-slip-truncated margin along the southwestern coast of Mexico that became active in the early Cenozoic and terminated volcanic arc activity as the Cayman-Motagua-Polochic strike-slip fault lengthened eastward (Karig *et al.*, 1978; Riller *et al.*, 1992; Herrmann *et al.*, 1994; Tardy *et al.*, 1994; Schaaf *et al.*, 1995). The 1100-km-wide Cayman trough pull-apart basin, bounded by strike-slip faults of the northern Caribbean plate boundary, opened as a result of large-scale, eastward strike-slip motion of the Caribbean plate (Rosencrantz *et al.*, 1988; Leroy *et al.*, 2000; Mann, 1999; Pindell *et al.*, 2006) (Fig. 1). In this interpretation, the 400–600-km-wide Chortis block is an intermediate continental block separating the larger continental area of northern Central America (Maya block) from Pacific subduction zones (Karig *et al.*, 1978; Gose, 1985; Schaaf *et al.*, 1995) (Figs. 1A and 1B).

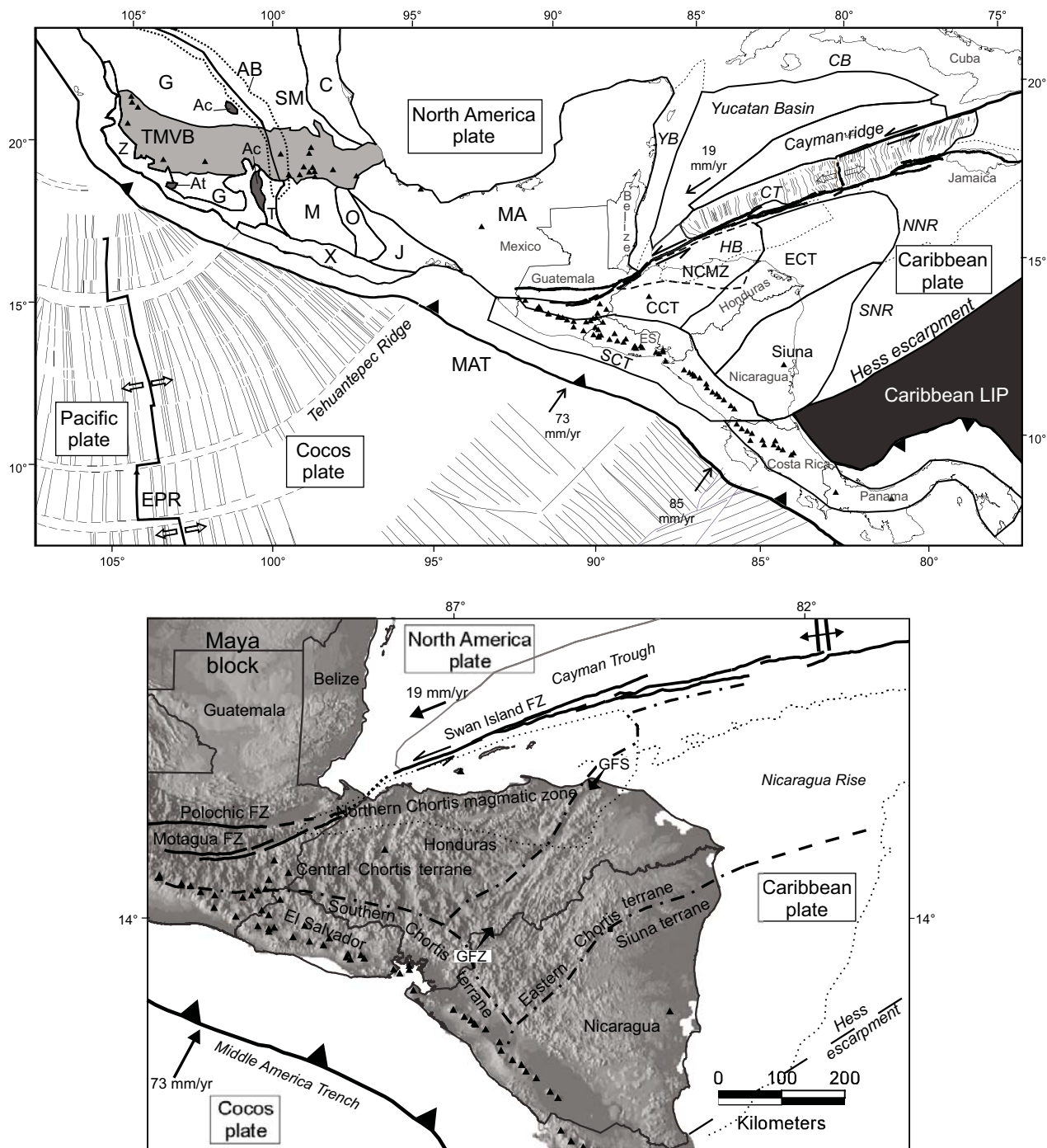


Figure 1. (A) Tectonic setting of northern Central America and southern Mexico showing the location and names of the Mexican terranes from previous workers and the Chortis terranes proposed in this paper. Chortis terrane abbreviations: CCT—Central Chortis terrane; ECT—Eastern Chortis terrane; SCT—Southern Chortis terrane; NCMZ—Northern Chortis metamorphic zone. Mexican terrane abbreviations: AB—Arperos basin, Mexico; Ac—Arcelia; At—Arteaga; C—Coahuila; G—Guerrero; J—Juarez; O—Oaxaca; M—Mixteca; MA—Maya block; SM—Sierra Madre; T—Teloloapan; X—Xolapa; and Z—Zihuatanejo. ES—El Salvador; TMVB—Trans-Mexican Volcanic Belt. Caribbean physiographic provinces shown as dashed lines and abbreviated as NNR—Northern Nicaraguan Rise; SNR—Southern Nicaraguan Rise; CB—Cuba; HB—Honduran borderlands; EPR—East Pacific Rise; MAT—Middle America trench; CT—Cayman trough; LIP—large igneous province; YB—Yucatan Basin. Triangles represent Quaternary volcanoes. Seafloor spreading anomalies shown for Pacific and Cocos plates and Cayman trough. Plate motions relative to a fixed Caribbean plate are from DeMets et al. (2000) and DeMets (2001). (B) Topographic map of northern Central America and the Chortis block showing physiographic expression of proposed terranes of the Chortis continental block. Note that structural grain and prominent faults like the Guayape fault system (GFS) are at right angles to the trend of the present-day Middle America trench and related volcanic arc (black triangles). Heavy dotted line offshore is 100 m bathymetric contour. FZ—fault zone.

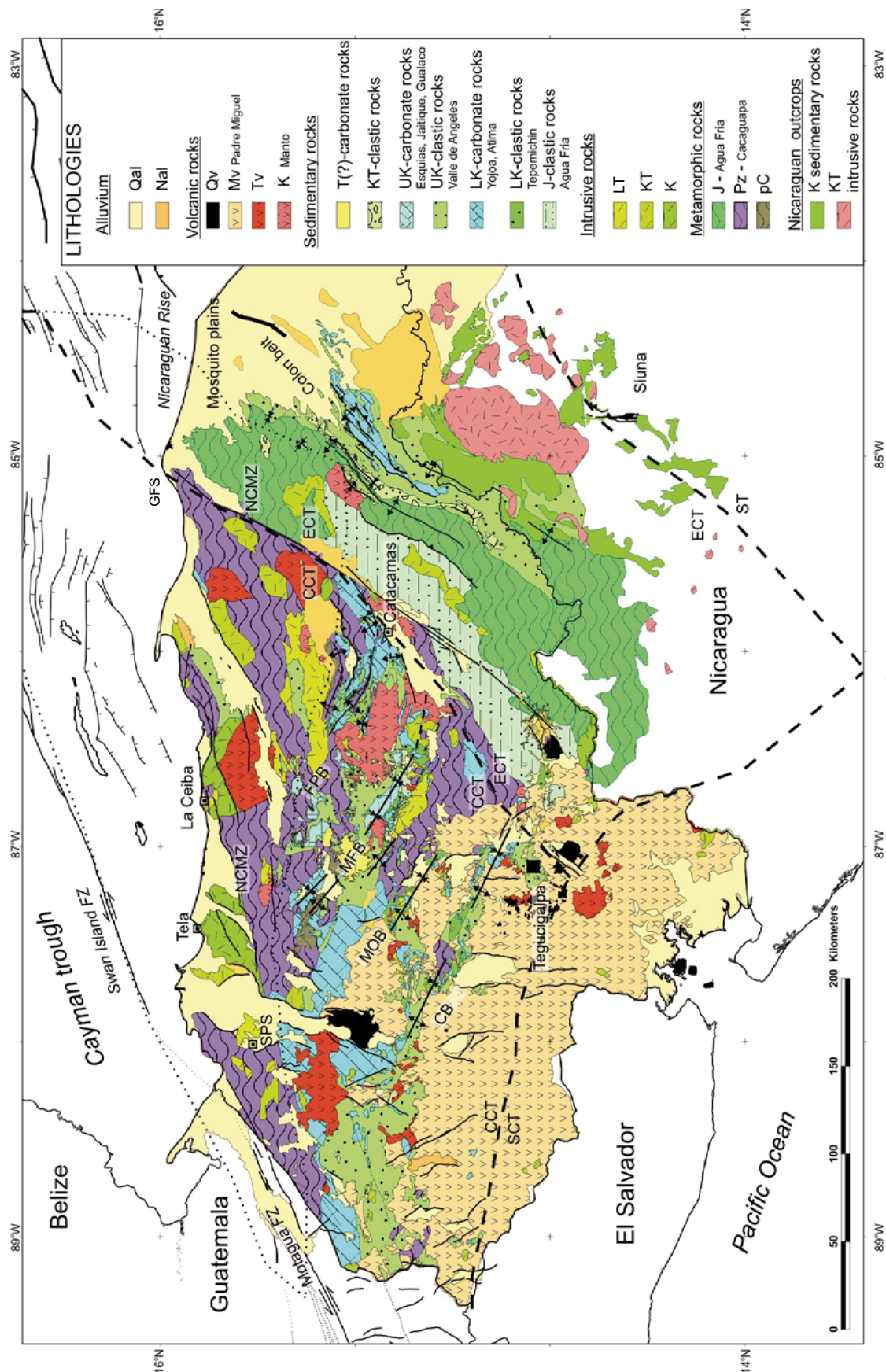


Figure 2. Geologic map of Honduras modified from national geologic compilation map by Kozuch (1991) combined with geologic map of northeastern Nicaragua modified from national geologic compilation map by the Instituto Nicaragüense de Estudios Territoriales (INETER, 1995). Dashed lines indicate terrane boundaries proposed in this paper using geologic and magnetic data. Paleozoic and Grenville-age basement occurs on the Central Chortis terrane. East of the Guayape fault system (GFS), mapping of the Eastern Chortis terrane reveals a basement composed of Jurassic Agua Fria strata and metasedimentary basement. Extensive Miocene ignimbrite deposits blanket much of western and southern Honduras and the boundary between the Central and Southern Chortis terranes. Lithologic abbreviations: Qal—Quaternary alluvium; Nal—Neogene alluvium; Qv—Quaternary volcanic rocks; Mv—Miocene volcanic rocks; Tv—Tertiary volcanic rocks; T—Tertiary; K—Cretaceous; KT—Cretaceous-Tertiary; UK—Upper Cretaceous; LK—Lower Cretaceous; J—Jurassic; IT—lower Tertiary; Pz—Paleozoic; and pC—Precambrian. CB—Comayagua belt; CCT—Central Chortis terrane; ECT—Eastern Chortis terrane; FPB—Frey Pedro belt; FZ—fault zone; MFB—Montaña La Flor belt; MOB—Minas de Oro belt; NCMZ—Northern Chortis magmatic zone; SCT—Southern Chortis terrane; SPS—San Pedro Sula; ST—Siuna terrane.

Pacific-Derived Model for Chortis

More recent mapping and isotopic dating based mainly in southern Mexico has led to a second, entirely different tectonic interpretation in which the Chortis block originates in the eastern Pacific Ocean and is ~700–800 km south of the truncated continental margin of southwestern Mexico during the Eocene (Keppie and Morán-Zenteno, 2005) (Fig. 1A). In this model, Chortis is far-traveled (1100 km) from 45 to 0 Ma and experiences ~40° of large-scale rotation in a clockwise sense. This sense of rotation is contrary to the overall, post-Cretaceous counterclockwise rotations measured by Gose and Swartz (1977) and Gose (1985) in rocks from Honduras.

In the Pacific-derived model, translation also occurs on the Cayman-Motagua-Polochic fault system, but the strike-slip fault is strongly concave southward about a distant pole of rotation in the southern hemisphere (cf. MacDonald, 1976, for a similar reconstruction). This fault geometry causes the Chortis block to travel in a northeastward arc from its original position far out in the eastern Pacific Ocean. In this model, a several hundred kilometer-wide zone of the southwestern margin of Mexico is removed by subduction erosion since the early Eocene. Continental rocks of southwestern Mexico and northern Central America are subject to the influence of subducting fracture zone ridges and other bathymetric highs that control the cessation and resumption of arc volcanism (Keppie and Morán-Zenteno, 2005).

Fixist Model for Chortis

A third type of model proposes that the Chortis block has occupied roughly its same position relative to southern Mexico and the rest of the Caribbean since Late Jurassic rifting between North and South America (Marton and Buffler, 1994; James, 2006) (Fig. 1A). This model is based on the interpretation of structural lineaments on the Chortis block and in surrounding areas that are inferred to be fossil rift and transform structures formed by the separation of North and South America during Late Jurassic time (James, 2006). The model also assumes that strike-slip displacement along the Motagua-Polochic-Cayman trough system (or between the Maya and Chortis blocks in Figs. 1A and 1B) is restricted to ~170 km of left-lateral offset.

OBJECTIVES OF THIS PAPER

In 2002, the Dirección General de Minas e Hidrocarburos of Honduras donated to the University of Texas at Austin the results of the national aeromagnetic survey of Honduras that was acquired by the Honduran government in 1985 as part of a countrywide program of exploration for minerals and petroleum (Dirección General de Minas e Hidrocarburos, 1985) (Fig. 3). These data provide an ideal tool for examining the distribution and lithologic-structural grain of the crystalline basement rocks of the sparsely mapped Chortis block because the source for most of the observed magnetic anomalies lies within the basement rocks, and the effects of the overlying sedimentary cover are essentially absent. The aeromagnetic data is particularly valuable for inferring the regional

trends and continuity of understudied basement rocks of the heavily vegetated and mountainous areas that characterize most parts of Honduras (cf. Fakundiny, 1970; Manton, 1996) (Fig. 2). Other basement rocks are buried beneath late Cenozoic sedimentary rocks of the coastal plain of eastern Honduras (Mills and Barton, 1996), lie beneath carbonate banks of the offshore Nicaraguan Rise (Arden, 1975), and are obscured by extensive volcanic and tuffaceous rocks related to the Miocene Central American volcanic arc in western Honduras (Jordan et al., this volume).

Our objective is to use this information to better constrain the trends of basement rocks making up the Chortis block, which has been inferred by most previous workers to be a single, monolithic basement block (cf. Dengo, 1985; Case et al., 1990; Pindell and Barrett, 1990).

We have defined tectonic terranes on the Chortis block following the criteria of Howell et al. (1985, p. 4)¹:

A tectonostratigraphic terrane is a fault-bounded package of rocks of regional extent characterized by a geologic history which differs from that of neighboring terranes. Terranes may be characterized internally by a distinctive stratigraphy, but in some cases a metamorphic or tectonic overprint is the most distinctive characteristic. In cases where juxtaposed terranes possess coeval strata, one must demonstrate different and unrelated geologic histories as well as the absence of intermediate lithofacies that might link the two terranes. In general, the basic characteristic of terranes is that the present spatial relations are not compatible with the inferred geologic histories. As additional geologic, paleontologic, and geophysical data are accumulated, terrane boundaries and classification can be modified and their nomenclature revised.

Our focus is the Chortis block of northern Central America, previously described by Case et al. (1984) as a “superterrane,” or amalgamation of crustal terranes, although these authors made no attempt to define the component terranes or establish their boundaries within the Chortis block. As pointed out by Williams and Hatcher (1982), tectonic terranes are assumed to be “guilty” or “suspect” until their “innocence” can be proven by establishing their origin, history, and relation to adjacent terranes. Due to the

¹Howell et al. (1985) define a terrane as a fault-bounded geologic entity or fragment that is characterized by a distinctive geologic history that differs markedly from that of adjacent terranes. Our field observations across the Chortis block reveal regions with markedly distinct geologic histories in the north, east, and south, reflected in differing stratigraphy and in the nature and grade of exposed basement rock. The extent to which these regions of the Chortis block are fault-bounded is uncertain because <15% of the Chortis block has been mapped at a scale of 1:50,000 or smaller. Therefore, instead of using field methods to delineate the boundaries of these distinct regions, we have employed a geophysical method that reflects the magnetic property of the basement rocks in each region. The correlation of magnetic property to the distinct geology of each region as described in this paper suggests that this approach is sound. The aeromagnetic survey we use encompasses a large part of the Chortis block, thereby allowing us to infer the boundaries of regions remote, inaccessible, or buried beneath young volcanic or alluvial strata. Therefore, the terranes we propose for the Chortis block are regions characterized by a distinctive geologic history that differs markedly from that of adjacent terranes that are in part fault-bounded and whose boundaries have been delineated on the basis of the magnetic properties of the basement rock.

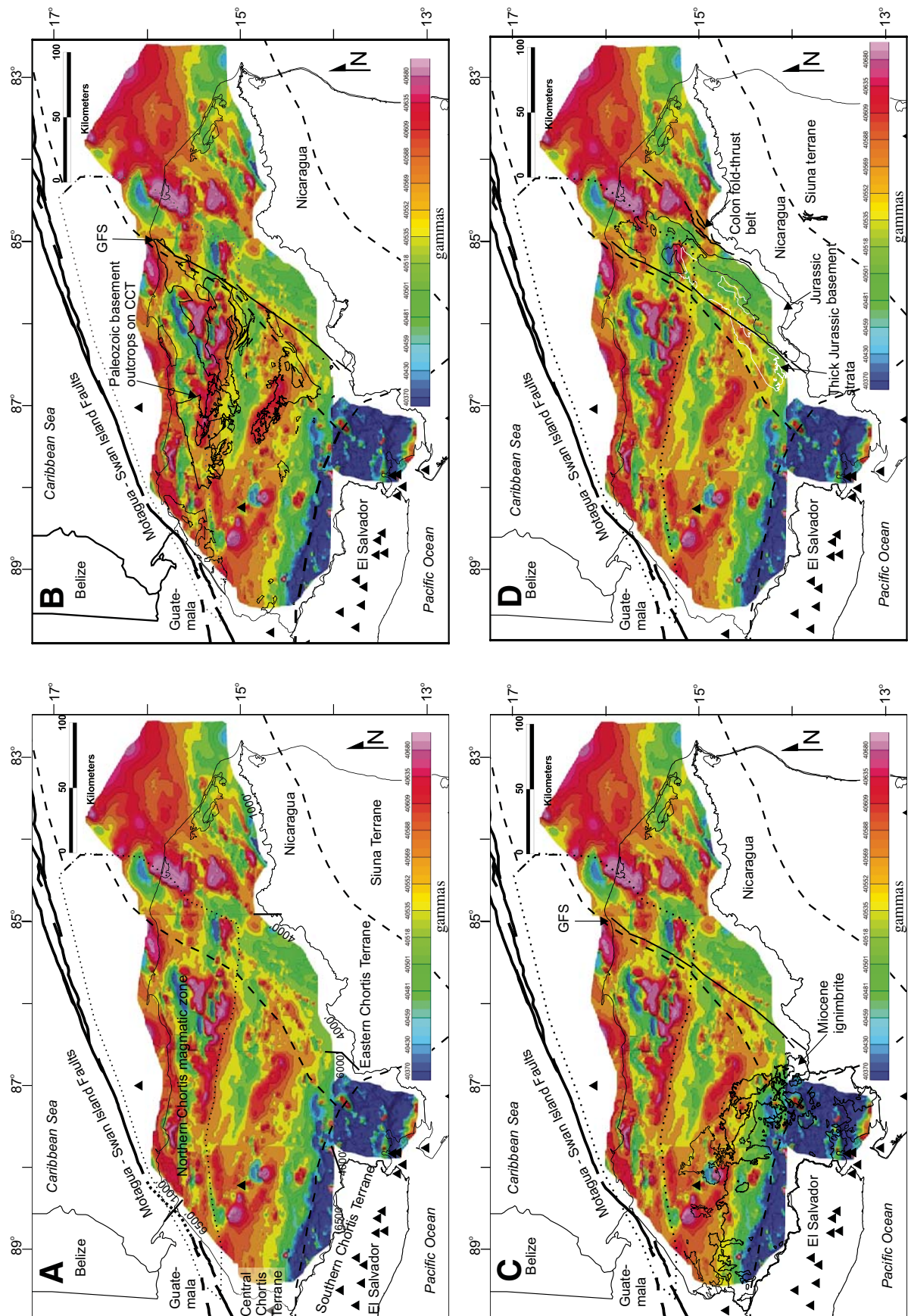


Figure 3. Composite total magnetic intensity anomaly map of Honduras (A) showing a higher but uniformly west-northwest-trending magnetic fabric across central Honduras (Central Chortis terrane); (B) a low-intensity region in southwestern Honduras (Southern Chortis terrane); (C) and a northeast-trending magnetic fabric in eastern Honduras (Eastern Chortis terrane) (D). The highly variable magnetic anomalies in northern Honduras (northern part of the Central Chortis terrane) are indicated by dotted outline. Quaternary volcanoes of the Central America volcanic arc are shown as black triangles. Key outcrop areas from the geologic map in Figure 2 are highlighted to show the geologic origins of the aeromagnetic anomalies. CCT—Central Chortis terrane; GFZ—Guayape Fault system.

sparse mapping of Honduras, the terranes we propose are “suspect” and subject to revision with improved mapping and dating. We have split the Chortis block into three terranes on the basis of aeromagnetic, geologic, and age isotopic observations until second-order analyses make the identity and genetic history of the pieces sufficiently well known to begin the “lumping” process. These proposed terranes and their boundaries should provide future workers a framework to plan more detailed mapping and dating studies.

Our objective is to characterize these suspect terranes and to identify the boundaries between these regions. To accomplish this, we utilize the regional aeromagnetic survey—a single geophysical data set that spans the regions of observed differing geology. We combine the aeromagnetic data with other geologic data compiled in this paper to (1) determine if this magnetic data can be correlated to the distribution of geologic features, and (2) determine if the magnetic data along with geologic data can be used to delineate the boundaries of these regions. Following these first two steps, using aeromagnetic and geologic data, we place the Chortis terranes in a more regional tectonic scheme by matching them with comparable tectonic terranes proposed by previous workers in southwestern Mexico (Fig. 1A).

GEOLOGIC AND TECTONIC SETTING OF THE CHORTIS BLOCK

The continental crust of the Chortis block forms the present-day northwest corner of the Caribbean plate, a plate otherwise composed of oceanic, oceanic plateau, or arc crust (Case et al., 1990) (Fig. 1). Subduction of the Cocos plate and its pre-Miocene predecessor, the Farallon plate, beneath the Chortis block (Caribbean plate) has produced the Middle America trench, a forearc basin, and the modern Central America volcanic arc (von Huene et al., 1980; Ranero et al., 2000) (Fig. 1).

Continental crustal rocks of northern Central America straddle the North America–Caribbean left-lateral, strike-slip plate margin, manifested by the prominent, arcuate-shaped fault valleys of the Motagua–Polochic fault strike-slip fault system in Guatemala (Figs. 1 and 2). Internal deformation of the northern and western parts of the Chortis block adjacent to these plate boundary faults occurs at rates of a few millimeters per year (DeMets et al., 2000; DeMets et al., this volume) has led to the formation of an east-west belt of about a dozen small, north-trending rifts of late Miocene to Holocene age (Burkart and Self, 1985; Guzman-Speziale, 2001; Rogers and Mann, this volume) (Figs. 1A and 1B). North of the strike-slip plate boundary, left-lateral motion of the North America–Caribbean plate is taken up by post-middle Miocene shortening, strike-slip faulting, and transpression in northern Guatemala (Guzman-Speziale and Meneses-Rocha, 2000). The southern and eastern boundaries of the Chortis block are loosely defined to be parallel to the Nicaragua–Costa Rica border and north of the Hess Escarpment in the Caribbean Sea (Dengo, 1985) (Fig. 2). GPS studies in these areas show that eastern Honduras is a stable region moving as part of the larger Caribbean plate (DeMets et al., this volume).

Geology of Central Honduras

Paleozoic continental crust (Cacaguapa Group of Fakundiny, 1970) is usually cited as the most common exposed basement type of the central Chortis block (cf. Horne et al., 1976a; Donnelly et al., 1990; Sedlock et al., 1993; Gordon 1993; Burkart, 1994) (Fig. 2). However, the oldest dated basement of Grenville age (1.0 Ga orthogneiss) has been found in several small inliers within larger areas of known or presumed Paleozoic crust in central and northern Honduras (Manton, 1996) (Fig. 2). In central Honduras, the Paleozoic gneiss and schist basement is overlain by mixed carbonate and clastic Cretaceous basins (Rogers et al., Chapter 5, this volume) (Fig. 4B). Both basement and overlying Cretaceous strata are deformed by folding and thrusting into elongate outcrops representing inverted Albian to Aptian-age intra-arc rifts (Rogers et al., Chapter 5, this volume) (Fig. 2).

Geology of Northern Honduras

In northern Honduras, basement rocks consist of highly sheared, high-grade gneiss of Paleozoic to Precambrian age, felsic batholiths of Late Cretaceous and Early Tertiary age, and Cretaceous age metasedimentary strata (Horne et al., 1976b; Manton, 1987; Manton and Manton, 1999; Avé Lallemant and Gordon, 1999) (Fig. 2). The age and degree of deformation of these rocks contrasts with the low-grade phyllite and schist of the Cacaguapa Group of central Honduras (Fakundiny, 1970) (Fig. 2).

Geology of Eastern Honduras and Nicaragua

Exposures of crystalline basement of the Chortis block have not been reported from eastern Honduras (Fig. 2). Metasedimentary basement outcrops in the region southeast of the Guayape fault system have been correlated with unmetamorphosed lithologies of the Jurassic Agua Fria Formation found northeast of the Guayape fault system (Viland et al., 1996; Rogers et al., Chapter 6, this volume). Folds and southeast-dipping reverse faults affecting Cretaceous carbonate and clastic rocks of the northeast-trending Colon fold-thrust belt parallel the northeast-trending fabric that is prominent on topographic maps and satellite imagery (Rogers et al., Chapter 6, this volume) (Figs. 2 and 4C). Most of our proposed boundary between the Central and Eastern Honduras follows the Guayape strike-slip fault that partially forms the western extent of the Jurassic Agua Fria metasedimentary basement type of eastern Honduras (Gordon and Muehlberger, 1994) (Fig. 2).

In northern Nicaragua, Venable (1994) used field, isotopic, and petrologic data to show that igneous basement rocks south of the mining area of Siuna, Nicaragua, are a Cretaceous island arc assemblage (including ultramafic cumulates and serpentinites) that formed on oceanic crust (Rogers et al., Chapter 6, this volume) (Fig. 2). She proposed that these rocks formed the Siuna terrane and the basement for much of Nicaragua.

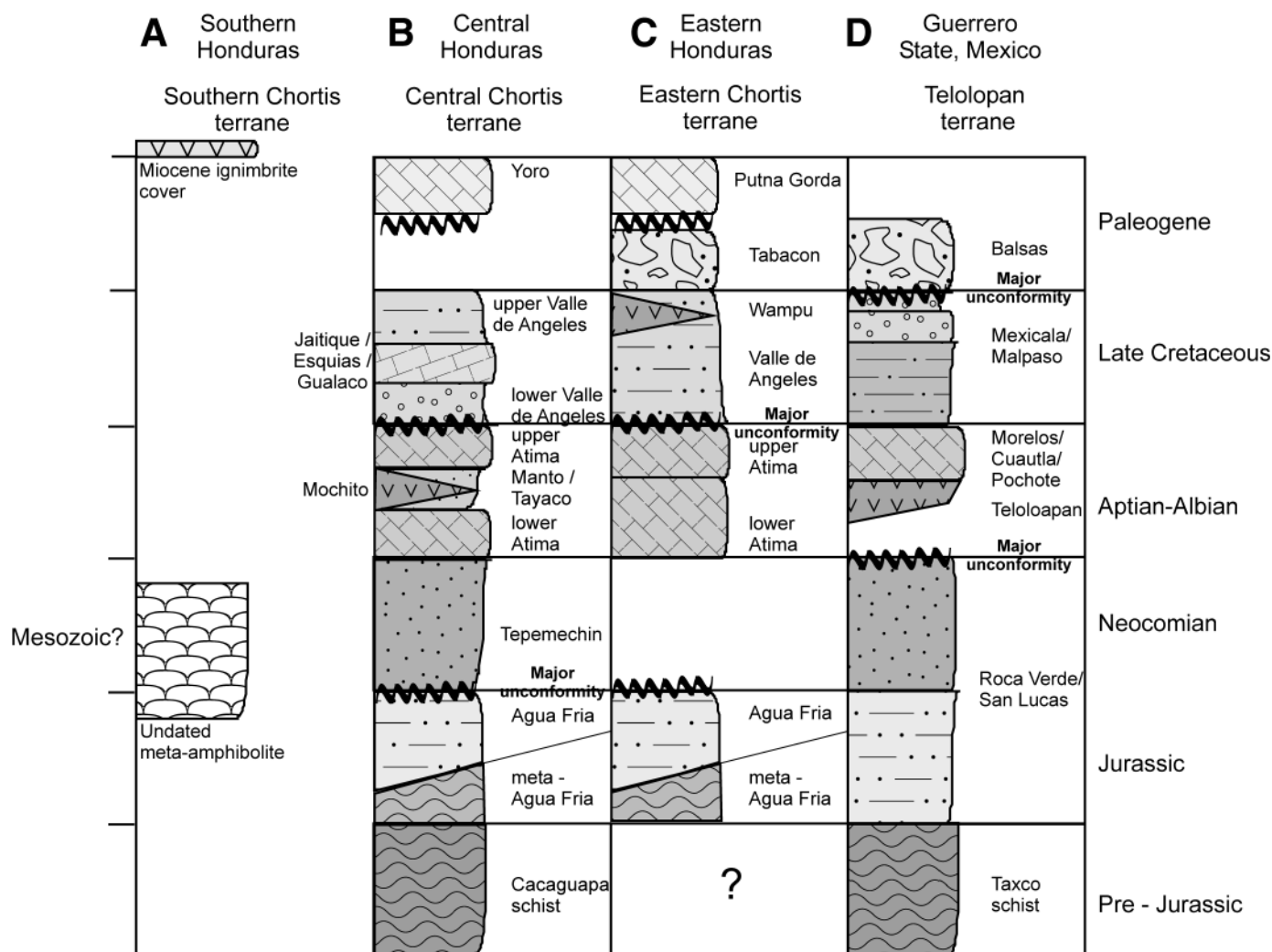


Figure 4. Regional correlation of late Mesozoic and Paleogene stratigraphy from Honduras and from the Guerrero state of southern Mexico. Data sources: Central Honduras terrane—Scott and Finch (1999); Kozuch (1991); and Rogers et al. (this volume, Chapter 5); eastern Honduras—Rogers et al. (this volume, Chapter 6); Guerrero state of Mexico—Cabral-Cano et al. (2000), Johnson et al. (1991), Cerca et al. (2006), and Lang et al. (1996). See text for discussion.

Geology of Southern Honduras

The distribution of Paleozoic rocks in southern and western Honduras is unknown because this region is blanketed by a thick cover, up to two kilometers, of Miocene volcanic strata produced by the Middle America volcanic arc (Williams and McBirney, 1969; Jordan et al., this volume) (Fig. 2). Geochemical data from modern lavas of the Central American volcanic arc show an absence of arc magma contamination or mixing with underlying Paleozoic continental crust of the type exposed in northern and Central Honduras (Carr et al., 2003). For this reason, Carr et al. (2003) infer that a post-Paleozoic, arc-type basement underlies the modern Middle America arc and areas extending an unknown distance north of the modern volcanic line. A single,

isolated exposure of highly deformed, metavolcanic amphibolite is exposed in southern Honduras along the gorge of the Rio Choluteca (Markey, 1995; Harwood, 1993) (Figs. 2 and 4A). This single occurrence of older, higher-grade rocks provides our only window into the underlying basement geology of our proposed Southern Chortis terrane.

AEROMAGNETIC DATA FROM HONDURAS AND THE EASTERN NICARAGUAN RISE

The high-sensitivity aeromagnetic survey comprises ~41,400 line kilometers distributed across six survey altitudes zones (1000, 4000, 6000, and 6500 barometric feet), evident by discontinuities on merged data (downward continuation)

(Fig. 3). Traverse spacing was 3 km and 5 km with tie-line spacing of 12 km and 20 km. The survey utilized a Varian High-Sensitivity Magnetometer and data was processed to develop total magnetic intensity contour maps that were corrected for topography at time of production in 1985. The survey did not cover strips along international borders of Honduras with neighboring countries but included an area of the Caribbean Sea near northeastern Honduras (Fig. 3).

The aeromagnetic data were provided to us by the government of Honduras in the form of a large paper map with contoured values of magnetic data at a scale of 1:500,000. We scanned the map on a large-format scanner and digitized the total magnetic intensity values at 20 gamma (nanotesla) intervals (i.e., we digitized every fifth contour of those that were shown on the original paper map). These data were compiled on a computer to produce the color contoured map shown in Figure 3. This contour interval was judged sufficiently dense to accurately convey the regional-scale patterns that are described as follows. The aeromagnetic data and surface geologic map data were overlain and precisely geo-referenced using a geographic information system (GIS).

The aeromagnetic survey provides an ideal tool for examining the distribution and nature of the basement of the Chortis block because the source of the observed magnetic anomalies lies within the basement rocks and the influence of the sedimentary cover signature is insignificant. The aeromagnetic survey provides one of the best possible sources of regional data to constrain the type and structure of the basement character underlying the volcanic regions of southern Honduras and the alluvial or water-covered areas of eastern Honduras and the Nicaragua Rise (Fig. 2).

REGIONAL MAGNETIC TRENDS VISIBLE IN HONDURAN AEROMAGNETIC DATA

Four Distinctive Magnetic Provinces

Four magnetic provinces are apparent in Figure 3: (1) an extremely low, northwest-trending magnetic intensity region in southwest and southern Honduras adjacent to the Central American volcanic arc; (2) a higher-intensity area of west-northwest-trending magnetic lineations across central Honduras; (3) a high-intensity, northeast-trending, lineated magnetic fabric in eastern Honduras; and (4) an area of high-intensity but highly variable speckled magnetic anomalies in northern Honduras.

Correlation of three of these magnetic provinces of Honduras with three geologic provinces of distinctive basement rocks, crustal properties, geochemical variations, and tectonic histories forms the basis of dividing the Chortis superterrane of Case et al. (1990) into three component terranes (Fig. 3A; Tables 1 and 2). In the following, each of the proposed Chortis terranes is defined based on correlation of known surface geology compiled

in Figure 2 with the aeromagnetic nature of the terrane shown in Figure 3.

Central Chortis Terrane

Terrane Definition

The Central Chortis terrane contains broad, lineated magnetic gradients trending west-northwest across central Honduras (Fig. 3) that parallel the belts of deformed basement and overlying Cretaceous strata (Fig. 2). Topographic ridges of elevated basement rocks of mainly Paleozoic and Precambrian age form magnetic highs, while regions of thick accumulations of overlying Cretaceous strata produce bands of lower magnetic intensity (Rogers et al., Chapter 5, this volume) (Fig. 3B). Age dating of the Central Chortis terrane is compiled on Table 1 from many previous authors and plotted on the magnetic map in Figure 5 from 1 Ga (Grenville) to 55 Ma (Eocene) south of the Northern Chortis magmatic zone.

A prominent oroclinal bend in these east-west belts of older basement rocks is observed in the topography, the geology, and on the aeromagnetic map (Figs. 2 and 3). In Chapter 5 of this volume, Rogers et al., propose that oroclinal bending accompanied a phase of left-lateral shear along the Guayape fault during the Late Cretaceous.

The four structural belts of Honduras (Fig. 2), representing inverted Aptian-Albian rifts (Rogers et al., Chapter 5, this volume), align well with the west-northwest-trending magnetic fabric in the Central Chortis terrane (Fig. 3). The late Neogene rifts of western Honduras are not discernable from magnetic intensity despite their prominent basin and range topographic expression (Figs. 1B and 2). The abrupt southern boundary of the Central Chortis terrane with the adjacent Southern Chortis terrane is easily discernible beneath the thick Miocene volcanic cover of western Honduras (Fig. 3).

We infer the Central Chortis terrane represents the core of the continental Chortis superterrane (Case et al., 1990) with presently northwest-trending Cretaceous intra-arc basins deforming Paleozoic basement (Figs. 2 and 4). Structural highs are now occupied by outcrops of Paleozoic and Precambrian rocks as seen on the outcrop overlay in Figure 3B. This Paleozoic and Precambrian basement is the continental basement type of the Chortis block most commonly described by previous authors, such as Fakundiny (1970) and Simonson (1977), and summarized on Table 2.

Northern Chortis Magmatic Zone

The aeromagnetic character of the northern part of the Central Chortis terrane maintains the same west-northwest trend observed in the Central Chortis terrane but the trend is more variable and punctuated by pairs of magnetic highs and lows (dipoles) that align with mapped, subcircular exposures of felsic batholiths of Late Cretaceous and early Tertiary age (Horne et al., 1976a, 1976b; Manton, 1996) (Figs. 2 and 3, Table 1). A belt of Late Cretaceous to early Tertiary magmatism occurs in

TABLE 1. COMPILATION OF CHORTIS BLOCK RADIOMETRIC DATA (cf. FIG. 5)

Sample	System	Mineral	Age (Ma)	Source and comments
<u>Northern Chortis metamorphic zone—basement rocks</u>				
Banaderos metagneous rock	Rb/Sr	wr	720 ± 260	Horne et al., 1976a
Quebrada Seca metagneous rock	Rb/Sr	wr	305 ± 12	Horne et al., 1976a
Adamellite gneiss	Rb/Sr	wr	230–203	Sedlock et al., 1993
Adamellite gneiss	Rb/Sr	wr	150–125	Sedlock et al., 1993
Chamelecon granodiorite	K/Ar	p	224.0 ± 17	MMAJ, 1980a—metagneous?
<u>Northern Chortis metamorphic zone—igneous rocks</u>				
San Marcos granodiorite	Rb/Sr	wr	150.0 ± 13	Horne et al., 1976a
Tela augite tonalite	K/Ar	h	93.3 ± 1.9	Horne et al., 1976b
Rio Jalan granite	U/Pb	z	92.7	Manton and Manton, 1984
Granodiorite	K/Ar		89.0 ± 4	MMAJ, 1980a
Chamelecon diorite	K/Ar		86.3 ± 3.7	MMAJ, 1980a
Rio Cangrejal gneiss	Rb/Sr	wr, b	83.5	Manton and Manton, 1984—sheared pluton
Rio Cangrejal gneiss	U/Pb	z	79.8	Manton and Manton, 1984—crystallization age
El Carbon granodiorite	U/Pb	z	81.0 ± 0.1	Manton and Manton, 1984—crystallization age
El Carbon granodiorite	U/Pb	z	76.3	Manton and Manton, 1984—metamorphic age
El Carbon granodiorite	Rb/Sr	k, p	68.9	Manton and Manton, 1984—metamorphic age
Tela augite tonalite	K/Ar	b	80.5 ± 1.5	Horne et al., 1976b
Trujillo granodiorite	Rb/Sr	wr	80.0	Manton and Manton, 1984
Tela augite tonalite	K/Ar	b	73.9 ± 1.5	Horne et al., 1976b
Piedras Negras tonalite	K/Ar	h	72.2 ± 1.5	Horne et al., 1976b
Mezapa tonalite	K/Ar	b	71.8 ± 1.4	Horne et al., 1976b
Las Mangas tonalite	K/Ar	m	57.3 ± 1.1	Horne et al., 1976b
Piedras Negras tonalite	K/Ar	b	56.8 ± 1.1	Horne et al., 1976b
Rio Cangrejal granodiorite	Rb/Sr	b	54.4	Manton and Manton, 1984
Sula Valley granodiorite	Rb/Sr	b	38.5	Manton and Manton, 1984
San Pedro Sula granodiorite	K/Ar	b	35.9 ± 0.7	Horne et al., 1976b
Confadia granite	Rb/Sr	b	35.0	Manton and Manton, 1984*
Banderos granodiorite	Rb/Sr	wr, b	30.8	Manton and Manton, 1984
Banderos granodiorite	U/Pb	z	28.9	Manton and Manton, 1984
<u>Central Chortis terrane—basement rocks</u>				
Orthogneiss	U/Pb	z	1000	Manton, 1996
Mica schist La Arada	K/Ar	b	222.0 ± 8	MMAJ, 1980b
<u>Central Chortis terrane—igneous rocks</u>				
San Andres dacite	U/Pb	z	124.0 ± 2	Drobe and Cann, 2000
San Ignacio granodiorite	K/Ar	h	122.7 ± 2.5	Emmet, 1983
San Ignacio granodiorite		b	117.0 ± 1.9	Emmet, 1983
San Ignacio adamellite	K/Ar	b	114.4 ± 1.7	Horne et al., 1976b
HN-149, basalt	K/Ar	wr	86.30 ± 3.8	ENEE, 1987
Gabbro dike	K/Ar	h	79.6 ± 1.6	Emmet, 1983
HN-205, diorite/basalt	K/Ar	wr	62.20 ± 2.6	ENEE, 1987
Minas de Oro granodiorite	K/Ar	b	60.6 ± 1.3	Horne et al., 1976b
Minas de Oro granodiorite	K/Ar	b	59.30 ± 0.9	Emmet, 1983
Minas de Oro granodiorite		h	55.00 ± 1.1	Emmet, 1983
San Francisco dacite pluton	K/Ar	b	58.6 ± 0.7	Horne et al., 1976b
<u>Eastern Chortis terrane—igneous rocks</u>				
Dipilto granite	Rb/Sr	wr	140.0 ± 15	Donnelly, et al., 1990
Carrizal diorite	Rb/Sr	k, b	118.0 ± 2	Manton and Manton, 1984
Wampu basaltic andesite	K/Ar	p	80.7 ± 4.3	Weiland et al., 1992
Wampu basaltic andesite	K/Ar	p	70.4 ± 3.4	Weiland et al., 1992
<u>Siuna terrane—igneous rocks</u>				
Andesite	Ar/Ar	h	75.62 ± 1.3	Venable, 1994
Diorite	Ar/Ar	b	59.88 ± 0.47	Venable, 1994
Note: Abbreviations: b—biotite; h—hornblende; z—zircon; p—plagioclase; k—potassium feldspar; wr—whole rock; ENEE—Empresa Nacional de Energia Eléctrica; MMAJ—Mineral Mining Agency of Japan. *Initial $^{87}\text{Sr}/^{86}\text{Sr} = 0.7040$ assumed.				

TABLE 2. SUMMARY OF GEOLOGIC CHARACTERISTICS OF THE CHORTIS TERRANES AND REGIONS (cf. FIGS. 3 AND 4)

Name	Type	Crustal Component	Defining characteristics	Exposed boundaries	Age	Origin
Central Chortis terrane (this study)	Continental fragment	Crystalline continental crust	Paleozoic basement, Cretaceous overlap sequence	Motagua-Polochic fault	Grenville-Paleozoic	Dispersed from nuclear Mexico
Eastern Chortis terrane (this study)	Continental margin fragment	Attenuated continental crust?	Meta-Jurassic basement, Jurassic strata, Cretaceous overlap sequence	Guayape fault, extent of Jurassic strata	Exposed Jurassic, inferred Paleozoic	Attenuated Central Chortis, dispersed from Mexico
Southern Chortis terrane (this study)	Volcanic arc fragment	Island arc built on oceanic crust	Magnetic, Tertiary volcanic overlap, inferred island arc basement	Inferred buried suture beneath Miocene ignimbrites	Inferred Jurassic-early Cretaceous	Accreted to Central Chortis, dispersed from Mexico
Siuna terrane (Venable, 1994)	Volcanic arc fragment	Island arc built on oceanic crust	Cretaceous volcanic basement, noncontinental isotopic signature, Tertiary volcanic overlap	Siuna suture	Early Cretaceous	Accreted to Eastern Chortis terranes
Northern Chortis metamorphic zone (this study)	Metamorphic	Overprinted continent and continental margin	Magnetic, high-grade metamorphic basement, Late Cretaceous-early Triassic plutons, Cretaceous overlap sequence metamorphosed	Shear zones, generally unmapped	Overprinted Grenville-Paleozoic	Metamorphic overprint of Central and Eastern Chortis terranes

a broad, 100-km-wide swath across northern Honduras (Horne et al., 1976a, 1976b; Kozuch, 1991) and is associated with a basement of sheared, high-grade gneiss (Manton, 1996; Manton and Manton, 1999) (Fig. 5, Tables 1 and 2). The concentrated belt of Late Cretaceous to early Tertiary magmatic activity on the northern part of the Central Chortis terrane suggests that the variability of the observed magnetic signature results from superimposed magmatic activity that we refer to here as the Northern Chortis magmatic zone. We do not distinguish this region as a separate terrane because its basement crustal types and isotopic ages are similar to the Central Honduras terrane to the south (Horne et al., 1976a, 1976b; Manton, 1996) (Fig. 5). The distinctive speckled aeromagnetic signature seen on the aeromagnetic map in Figure 3 is clearly related to intrusions within the Northern Chortis magmatic zone and therefore a local thermal and intrusive event affecting only the northern part of the Central Chortis terrane.

The variable character of the magnetic field of the Northern Chortis magmatic zone continues into the Eastern Chortis terrane where it is characterized by large high-low magnetic dipoles occurring beneath the northwestern part of the Mosquitia Plain and the Nicaragua Rise, suggesting buried intrusive bodies in the subsurface. Arden (1975) describes oil industry wells from the Nicaraguan Rise that encountered plutonic rocks of Late Cretaceous and early Cenozoic age that are unconformably overlain by Cenozoic carbonate banks of the Nicaraguan Rise.

The uncertain boundaries of the overprinting effect of the magmatic province across the northern margin of the Central Chortis block is indicated by the dotted lines in Figure 3. As drawn, the Northern Chortis magmatic zone also overprints the Guayape fault system shown in Figure 3 in a manner that indicates the Late Cretaceous pulse of magmatic overprinting

post-dates significant lateral offset along the fault (Gordon and Muehlberger, 1994). (Figs. 2 and 5).

Faulting Related to the Strike-Slip Plate Margin

Some of the variability of the magnetic field of the northern part of the Central Chortis terrane appears to have been produced by late Neogene transtensional deformation of the Honduras borderlands related to the North America–Caribbean strike-slip plate boundary (Rogers and Mann, this volume) (Fig. 2). This late tectonic control on the aeromagnetic signature is shown by the prominent magnetic low aligned with the active, alluvial-filled Aguan rift basin south of the northern coastal mountain range of Honduras (Figs. 2 and 3).

Eastern Chortis Terrane

Magnetic intensity of the Eastern Chortis terrane trends to the northeast (Fig. 3) and is confined mainly to the area east of the Guayape fault zone (Fig. 2). The regional magnetic fabric of the Eastern Chortis terrane is roughly perpendicular to the prominent west-northwest magnetic trends of the Central Chortis terrane (Fig. 3). The boundary between the Central and Eastern Chortis terranes is placed at the eastern extent of the crystalline basement exposure and coincides with the northern part of the Guayape fault. In this area, the pronounced northeast-trending topographic fabric of the Eastern Chortis terrane is well-reflected by the northeast orientation of outcrops and trends in the Colon fold-thrust belt of Late Cretaceous age (Rogers et al., Chapter 6, this volume) (Figs. 2 and 3D).

—In the area of the southern Guayape fault zone south of the town of Catacamas (Fig. 2), we propose that the boundary separating the Central and Eastern terranes does not coincide with

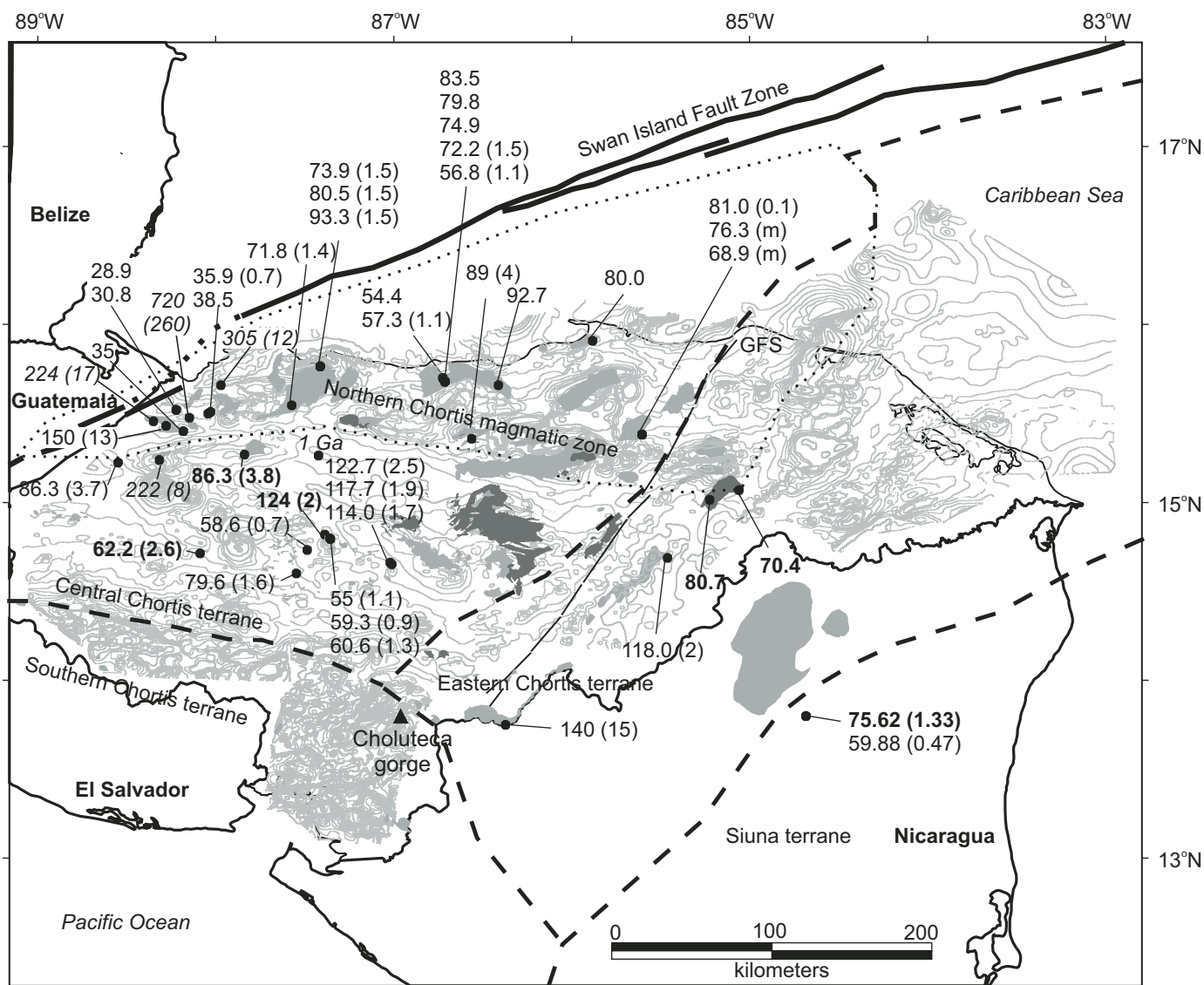


Figure 5. Compilation map of pre-Neogene reported radiometric ages of igneous rocks in Honduras and Nicaragua, in m.y.; error shown in parentheses (also in m.y.). Outcrop exposures of all plutonic rocks from the geologic map in Figure 2 are shown by light shading; outcrop exposures of all volcanic rocks are shown in dark shading (for simplicity, Neogene volcanic rocks are omitted). Note the concentration of Late Cretaceous to early Tertiary magmatic activity that coincides with the northern edge of the Central Chortis terrane. Bold numbers indicate reported radiometric dates for volcanic rocks; italics indicate dates from metamorphic rocks; (m) denotes radiometric dates for the deformation age of sheared igneous rocks. Gray background contours are magnetic intensity map of Honduras shown in Figure 3. Sources for all data are given in Table 1. Triangle—location of the Choluteca gorge exposure of basement rock in the Southern Chortis terrane; GFS—Guayape Fault system.

the Guayape fault but instead follows a prominent magnetic lineament west of the Guayape fault (Fig. 3). This magnetic lineament roughly coincides with the western margin of the Jurassic sedimentary basin (Rogers et al., Chapter 6, this volume). A small exposure of crystalline basement rock occurs adjacent to the southernmost segment of the Guayape fault zone, although it is not clear by what mechanism this area was exposed (Rogers, 1995).

We infer that the East Chortis terrane represents the rifted, southern margin of Chortis (North American plate) developed during the Jurassic opening of the North and South America plates (Table 2). The original Jurassic orientation of these rift-related faults was likely to be more east-west than their present strike because a large, Cenozoic counterclockwise rotation needs to be taken into account (Gose and Swartz, 1977; Gose, 1985). This implies that attenuated crystalline continental basement may exist beneath the exposed Jurassic metasedimentary rocks of the Eastern Chortis terrane.

We have no direct evidence for Jurassic age extensional faults, especially on the regional scale depicted by James (2006). Thickening of the Jurassic sedimentary units from a few tens of meters on the Central Chortis terrane to >2 km on the Eastern Chortis terrane lends some support to the interpretation by James (2006) that at least part of the Guayape fault may have originated as a Late Jurassic normal fault. Exposed basement of the Eastern Chortis terrane is limited to greenschist-grade phyllite and schist of the metamorphosed Jurassic-age Agua Fria Formation (Rogers et al., Chapter 6, this volume). The northeast magnetic trends of the Eastern Chortis terrane align well with the distribution of folded Jurassic rocks. These pronounced magnetic trends indicate that folds and faults are likely thick-skinned features that may also affect inferred crystalline basement rocks at depth beneath the outcrops.

Southern Chortis Terrane

The boundary between the Southern and the Central Chortis terranes appears on the magnetic map of Honduras as 200-km-long lineament across which an abrupt southward decrease in total magnetic intensity is exhibited (Fig. 3). The low but variable magnetic field in the Southern Chortis terrane results in an undulating pattern of magnetic highs and lows that do not correlate with Quaternary volcanic centers in the area (Fig. 2). The Southern Chortis terrane is overlain by up to 2 km of mid-Miocene pyroclastic strata of the Chortis block (Williams and McBirney, 1969; Jordan et al., this volume) (Fig. 2). A single known exposure of basement of the Southern Chortis terrane is a metavolcanic amphibolite mapped by Markey (1995) (Fig. 5).

The absence of the Paleozoic basement underlying the Southern Chortis terrane is indicated by the geochemistry of Quaternary volcanic lavas (Carr et al., 2003). We propose that post-Paleozoic arc or oceanic-type basement of the Southern Chortis terrane may extend to the southeast and underlie the offshore Cretaceous forearc basins (von Huene et al., 1980; Ranero et al., 2000) (Fig. 1).

Because the Southern Chortis terrane sharply abuts the southwestern edge of the continental Central Chortis terrane, the Southern Chortis terrane is inferred to be an accreted oceanic element to more inboard continental terranes (as is the case for the Guerrero terrane to nuclear Mexico) (Centeno-Garcia et al., 1993). Future work may identify more outcrops of the basement rock of the Southern Chortis terrane, which is now known only from one locality (Fig. 5).

Siuna Terrane

Adjacent to the Chortis terranes is the Siuna terrane (Venable 1994) in the area of the Siuna mining district in northern Nicaragua (Figs. 2 and 5). Because the area is in Nicaragua, it is not covered by the Honduras aeromagnetic map shown in Figure 3. At Siuna, exposures of thrustured serpentinite and associated ultramafic cumulates exhibit isotopic characteristics of oceanic crust (Rogers et al., Chapter 6, this volume) and is consistent with the interpretation that the Siuna terrane consists of an Early Cretaceous oceanic island arc developed on oceanic basement and accreted to the Eastern Chortis terrane in the Late Cretaceous (Venable, 1994). The Colon fold-thrust belt in eastern Honduras and the northern Nicaragua Rise developed in response to the suturing of the Siuna terrane to the eastern Chortis terrane in the Late Cretaceous (Rogers et al., Chapter 6, this volume).

This terrane exposed in northern Nicaragua is an exotic oceanic crustal element accreted to the Eastern Chortis terrane in the Late Cretaceous (Table 2). We infer that the close association of the Siuna terrane island arc with the Colon fold-thrust belt of the Eastern Chortis terrane indicates that both were deformed in the arc-continent collision between the Guerrero-Caribbean arc and the continental margin of the southern Chortis block (Rogers et al., Chapter 6, this volume).

Kesler et al. (1990) show that the continental blocks of Central America and Mexico display distinct clustering of lead isotopic ratios from samples taken from basement and overlying volcanic rocks. These lead values therefore provide a useful basis to distinguish among the complexly amalgamated terranes of Central America and Mexico. A plot of lead values is shown in Figure 6 for basement or volcanic samples from the following terranes: Central Chortis, Siuna, Eastern Chortis, Tertiary volcanic rocks in Nicaragua, the Maya block, and the Caribbean large igneous province (all sources of data are given in the figure caption and are discussed in detail by Venable (1994) and Rogers et al. (Chapter 6, this volume). The lead values of the Siuna terrane cluster outside the Caribbean large igneous province and indicate that the Siuna arc is not underlain by crust of the Caribbean large igneous province (Fig. 6). Instead, it is likely that the Caribbean arc system developed at the edge of the Caribbean large igneous province rather than directly above it. Other interesting correlations are that lead values from the Central and Eastern Chortis areas form distinct clusters and support our proposal that both areas occupy distinctive terranes.

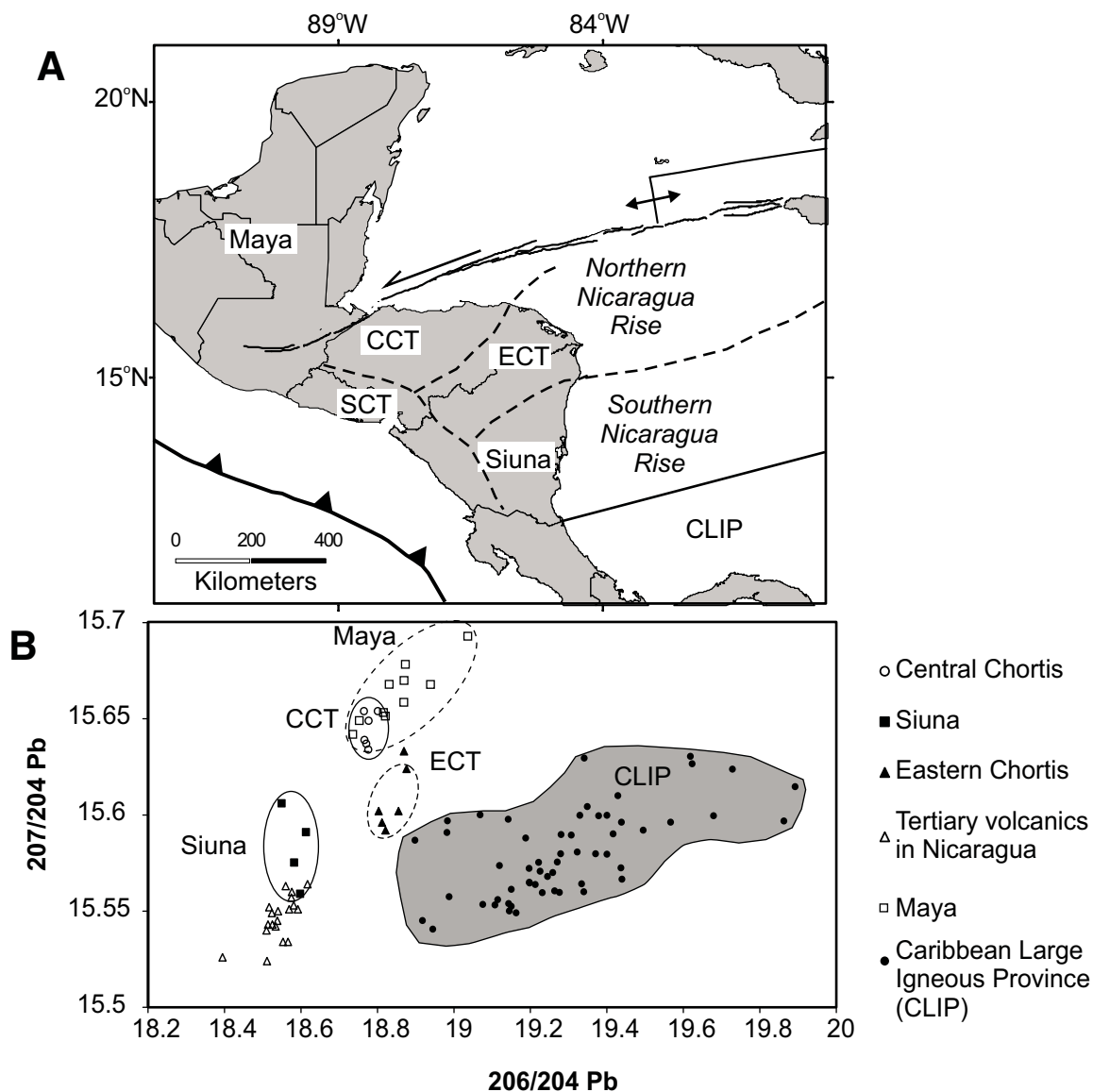


Figure 6. (A) Map showing Maya and Chortis blocks and proposed terranes of the Chortis block: CCT—Central Chortis terrane; ECT—Eastern Chortis terrane; SCT—Southern Chortis terrane; Siuna—Siuna terrane. Summary of lead isotopic data from Maya and Chortis continental blocks. (B) Comparison of common lead isotope data for Chortis block (including the Siuna terrane) with surrounding terranes and the Maya block. Note that lead data distinguishes the accreted Siuna terrane inferred to have originated as part of the Great Arc of the Caribbean. No lead data currently exists for the magmatic rocks along the northern part of the Central Chortis terrane. Lead data compiled from previous work by (Cumming and Kesler, 1976; Cumming et al. 1981); Sunblad et al. (1979); Sinton et al. (1997); Hauff et al. (2000); and Hoernle et al. (2002).

DISCUSSION

Defining Terranes on the Chortis Block

In order to define the Chortis terranes, we have applied the definition of a tectonic terrane as a “fault-bounded geologic entity or fragment that is characterized by a distinctive geologic history that differs markedly from that of adjacent terranes” (Howell et al., 1985, p. 4). Our terrane definitions are preliminary because of the large tracts of Honduras that remained unstudied or are known only at a reconnaissance level. However, the use of the aeromagnetic data supports our interpretations and shows that the Chortis block is a monolithic “superterrane” as envisioned by Case et al. (1984) and appears to be a composite of at least three terranes that have become welded together by Jurassic normal faulting or later strike-slip and shortening events along relatively narrow fault zones. Due to the sparse mapping of Honduras, the terranes we propose are “suspect” and subject to revision with improved mapping and dating. However, these proposed terranes and their boundaries should provide future workers a framework by which to plan more detailed mapping and dating studies.

PIERCING LINES AND OTHER FEATURES COMMON TO SOUTHERN MEXICO AND THE CHORTIS BLOCK

Many previous workers have speculated that the Chortis block originated along the southwestern margin of Mexico prior to Cenozoic translation from the North America plate to the Caribbean plate (Dengo, 1985; Ross and Scotese, 1988; Pindell and Barrett, 1990; Riller et al., 1992; Herrmann et al., 1994; Schaaf et al., 1995; Dickinson and Lawton, 2001). Placement of the Chortis along the southern Mexico margin by these workers has remained speculative because these reconstructions lacked piercing points or lines that could reconnect the rocks in Honduras with those in southwestern Mexico.

In this discussion, we introduce six geological and geophysical features to constrain the Late Cretaceous position of the Chortis block along the southwestern Mexican margin. Two features and three piercing lines are common to southern Mexico and the Chortis block: (1) Precambrian basement, (2) similar Mesozoic cover, (3) a north-trending mid-Cretaceous arc and geochemical trends, (4) north-trending Late Cretaceous structural belts, and (5) a north-trending common magnetic signature. (6) A fourth piercing line is the east-trending alignment of the eastern Honduras Colon fold belt with Late Cretaceous east-trending fold belt of southeastern Guatemala (Rogers et al., Chapter 6, this volume). Our best fit alignment of these features and piercing lines places the Chortis block along the southern margin of Mexico as displayed on the latest Cretaceous reconstruction in Figure 7A.

Common Feature 1 of Chortis and Southern Mexico

The Precambrian basement of Mexico is documented in the Oaxaca region of Mexico and is believed to represent the

southward continuation of North American Gondwana elements into Mexico (Ortega-Gutierrez et al., 1995; Dickinson and Lawton, 2001). The distribution of Precambrian basement exposures (Oaxaca terrane of Ortega-Gutierrez et al., 1995) in Mexico is shown on Figures 1A and 6. On the Chortis block, Manton (1996) reports Precambrian basement of 1.0 Ga in the Yoro region of Honduras (Fig. 7A, near the circled “3”). The reconstruction we propose Figure 7A juxtaposes these two areas of Precambrian crust in pre-Eocene times.

Common Feature 2 of Chortis and Southern Mexico

A remarkably similar sequence of Late Cretaceous clastic, marine sandstone and shale overlying an Early Cretaceous shallow-water carbonate platform rocks occurs both on the Chortis block and in the area of the Teloloapan terrane of Guerrero state in southern Mexico (Figs. 4 and 7A). In Honduras, Late Cretaceous marine sandstone and shale is represented by the Valle de Angeles Formation (Figs. 4B and 4C) (Mills et al., 1967; Wilson, 1974; Horne et al., 1974; Finch, 1981; Scott and Finch, 1999; Rogers et al., this volume, Chapters 5 and 6) and in Mexico by the Mexicala Formation (Johnson et al., 1991; Lang et al., 1996; Cabral-Cano et al., 2000; Cerca et al., 2004) (Fig. 4D). Mid-Cretaceous carbonate deposition resulted in deposition of the Morelos Formation in Mexico (e.g., Cabral-Cano et al., 2000) and the Atima Formation on the Chortis block (e.g., Finch, 1981). The continuity of Cretaceous strata across the Eastern and Central Chortis terranes and into Mexico constitutes an overlap assembly above the older Paleozoic and Precambrian terranes. The reconstruction in Figure 7A restores both areas of similar Cretaceous stratigraphy.

North-South Piercing Line 1 of Chortis and Southern Mexico

The Early Cretaceous and arc affinity of the Manto Formation on the Chortis block correlates with Early Cretaceous arc activity in southern Mexico (Rogers, et al., Chapter 5, this volume) (Figs. 2 and 7A). In Mexico, three separate Early Cretaceous arcs have been recognized and include the Arteaga, Arcelia, and Teloloapan (Mendoza and Suastegui, 2000). Rogers et al., (this volume, Chapter 5) compared the multi-elemental geochemical patterns of the Manto Formation volcanic rocks of Honduras with the reported values for the three Early Cretaceous volcanic arcs in Mexico and concluded that the geochemical data supports a direct correlation between the Teloloapan arc of Mexico and the Manto Formation of Honduras. The reconstruction in Figure 7 restores both areas of similar Early Cretaceous arc activity.

North-South Piercing Line 2 of Chortis and Southern Mexico

Late Cretaceous shortening, attributed to Laramide-age deformation in Mexico, is observed both in southern Mexico (Campa, 1985; Burkart et al., 1987; Lang et al., 1996; Cerca et al., 2004) and on the Chortis block (Horne et al., 1974; Donnelly et al., 1990; Rogers et al., Chapter 5, 2006). Our best-fit positioning of the Chortis block along the southern margin Mexico aligns the Late Cretaceous north-trending fold thrust belts of Mexico with the four Late Cretaceous structural belts in Honduras (Fig. 7).

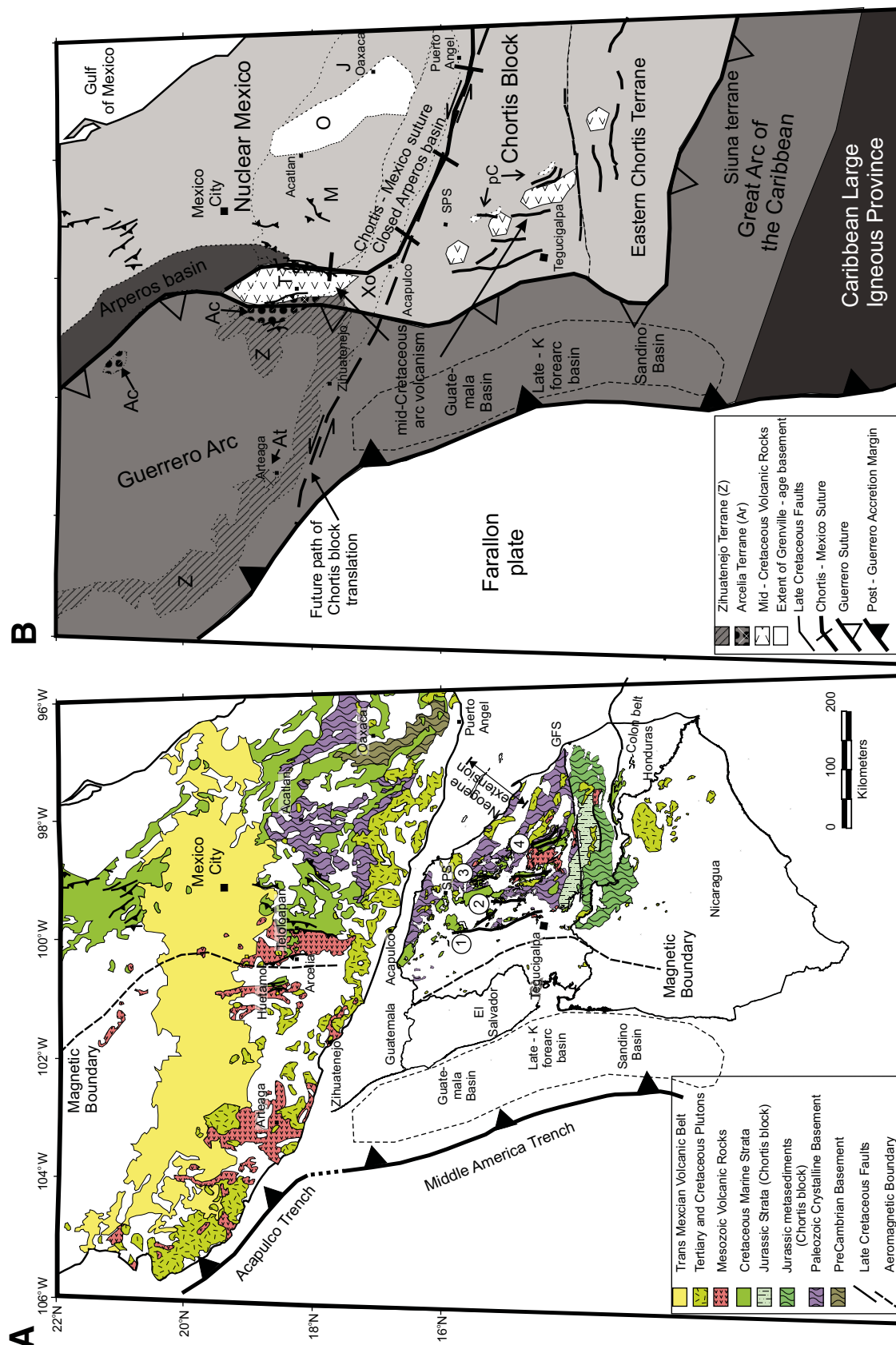


Figure 7. (A) Restoration of the Chortis block to a position adjacent to southwest Mexico in order to realign similar, pre-Tertiary structures and geologic units on the Chortis block and in southwestern Mexico. Key features in Honduras and southern Mexico: (1) a Cretaceous volcanic arc (cf. Rogers et al., Chapter 5, this volume); (2) a late Mesozoic stratigraphy (cf. stratigraphic columns in Fig. 4); (3) a Grenville-age basement exposure of Central Honduras terrane; (4) aligned rifts formed in late Cretaceous times (Rogers et al., Chapter 5, this volume); and (5) a distinct magnetic boundary between the accreted Guerrero terrane and nuclear Mexico and between the Southern and Central Chortis terranes boundaries (Fig. 8). Structural belts on the Chortis block are noted by circled numbers: 1—Comayagua belt; 2—Minas de Oro belt; 3—Montaña de la Flor belt; 4—Frey Pedro belt; abbreviations: SPS—San Pedro Sula; GFS—Guayape fault system. (B) Latest Cretaceous tectonic setting of the southwestern corner of the North American continent prior to the proposed eastward displacement of the Chortis block as part of the Caribbean plate. Following previous workers, we assume that the Guerrero arc and the Great arc of the Caribbean form a continuous, west and north-facing arc system that wraps around the continental promontory of southwestern North America. These large scale lithologic trends are supported by the aeromagnetic data shown in Figure 3. Ac—Arceia; At—Artega; J—Jurassic; K—Cretaceous; M—Miocene; O—Oligocene; pC—pre-Cambrian; SPS—San Pedro Sula; T—Tertiary; Xo—Xolopa.

North-South Piercing Line 3 of Chortis and Southern Mexico

Prominent magnetic expression boundaries exist on both the Chortis block (Fig. 3) and in western Mexico (Fig. 8) with the location of the main terrane boundaries shown on Figure 7. In Mexico, the prominent magnetic boundary separates the accreted oceanic and arc Guerrero terrane from the autochthonous continental terranes of nuclear Mexico (Campa and Coney, 1983; Dickinson and Lawton, 2001). A similar relationship is present on the Chortis block where a major terrane boundary separates the Southern Chortis terrane known from a single basement to be underlain by metaigneous (perhaps oceanic) rocks and the Central Chortis terranes known from outcrop studies to be underlain by continental rocks of Precambrian and Paleozoic age (Fig. 4).

East-West Piercing Line of Chortis and Southern Mexico

The Colon foldbelt of eastern Honduras represents the collisional response to the suturing of an island arc to the southern margin of North America in the late Cretaceous (Rogers et al., this volume, Chapter 6). We interpret this as the eastern extent of the same event that emplaced the Santa Cruz ophiolites and deformed the southern Guatemala in the late Cretaceous (Rosenfeld, 1981; Donnelly et al., 1990). Cenozoic translation of the Chortis to the southeast places the Colon belt to the east of its Guatemala counterpart.

Restoring the Chortis block along the southern margin of Mexico based on the best-fit alignment of features common to both regions results in the following configuration of the southwestern corner of North America prior to the eastward Cenozoic translation of the Chortis block that is summarized on the map in Figure 7B:

1. Continuity of the Chortis block and the autochthonous terranes of nuclear Mexico provides a common basement and platform for the deposition of late Mesozoic strata and its subsequent Late Cretaceous shortening (Fig. 7).
2. Continuity of the geochemically similar Cretaceous Telo-loapan and Manto arcs links a north-south-trending arc now offset left-laterally (Fig. 7).
3. An open western and southern margin along the southwest corner of North America provide a backstop for the Guerrero arc and the Caribbean arc to accrete. Caribbean arc components include the Guerrero terrane, the southern Chortis terrane, and the Siuna terrane (Fig. 7).

Correlation between Chortis Terranes and Terranes in Southwestern Mexico

Using the common features and piercing lines between the Chortis block and southwestern Mexico, the late Cretaceous alignment of the Chortis block along the southern Mexico margin corresponds with the alignment of similar Chortis and Mexican terranes (Fig. 7). In this position we propose a correlation between the Chortis terranes and the terranes in Mexico based on common type and age of basement and correlative events (i.e., stratigraphic, igneous and deformation). We utilize the terrane

names of Campa and Coney (1983) rather than the different set of terms introduced by Sedlock et al., (1993). We take into account the recent terrane revision of Keppie (2004) where his revision is relevant to our proposed correlations.

To facilitate discussion of the various crustal elements of the southern Mexican Cordillera, Central America and the Caribbean studied by many previous authors, Table 3 summarizes key crustal elements of all proposed terranes. This table lists the terrane nomenclature used in the discussion below in the left hand column and is consistent with labeling of terranes shown on the regional map in Figure 1A. The terranes are organized by affinity to crustal elements following the approach of Dickinson and Lawton (2001) and include some of the modifications by Keppie (2004).

Central Chortis Terrane—Del Sur Block (Mixteco-Oaxaca Terranes)

The Central Chortis terrane with its basement of Paleozoic and Grenville-age metamorphic rocks and thick continental crust is correlated with the Del Sur block (Dickinson and Lawton, 2001) composed of elements of neighboring Mixteca and Oaxaca terranes (Campa and Coney, 1983; Sedlock et al., 1993) and the southern Oaxaquia terrane of Keppie (2004) (Fig. 1). Similarities between Cretaceous sequences of the central Chortis terrane and those of the Mixteco terranes and the Teloloapan subterrane of the Guerrero terrane include: (1) early Cretaceous rifting, platform carbonate and terrestrially derived redbed deposition; (2) development of intra-arc volcanism; and (3) Laramide-age inversion of Mesozoic sequences (Lang et al., 1996; Mendoza and Suastegui, 2000; Cabral-Cano, et al., 2000; Cerca et al., 2004; Figure 4 in Rogers et al., Chapter 5, this volume).

The Eastern Chortis terrane with a Jurassic metasedimentary basement was controlled by opening and rifting between the Americas in early Jurassic time (cf. Pindell and Barrett, 1990). The little studied Juarez or Cuicateco terrane of Mexico with its Jurassic metasedimentary and metavolcanic basement (Sedlock et al. 1993) appears a likely genetic equivalent to the Eastern Chortis terrane (Fig. 1). Dickinson and Lawton (2001) note that the Juarez terrane likely formed during proto-Caribbean opening and prior to the early Cretaceous arrival of the Maya (Yucatan) block from the northern Gulf of Mexico (Marton and Buffler, 1994).

The Southern Chortis Terrane—Guerrero Terrane

We correlate the Southern Chortis terrane to the Mexican Guerrero composite terrane. The Mexican Guerrero terrane is a Jurassic-Cretaceous island arc built on oceanic crust accreted to the western margin of Mexico (Centeno-Garcia et al., 1993; Sedlock et al., 1993). Guerrero accretion to western Mexico is interpreted as a diachronous collision from north to south spanning much of Cretaceous time (Tardy et al., 1994; Dickinson and Lawton, 2001).

The aeromagnetic map of Mexico (North American Magnetic Anomaly Group, 2002) displays a strong contrast between nuclear

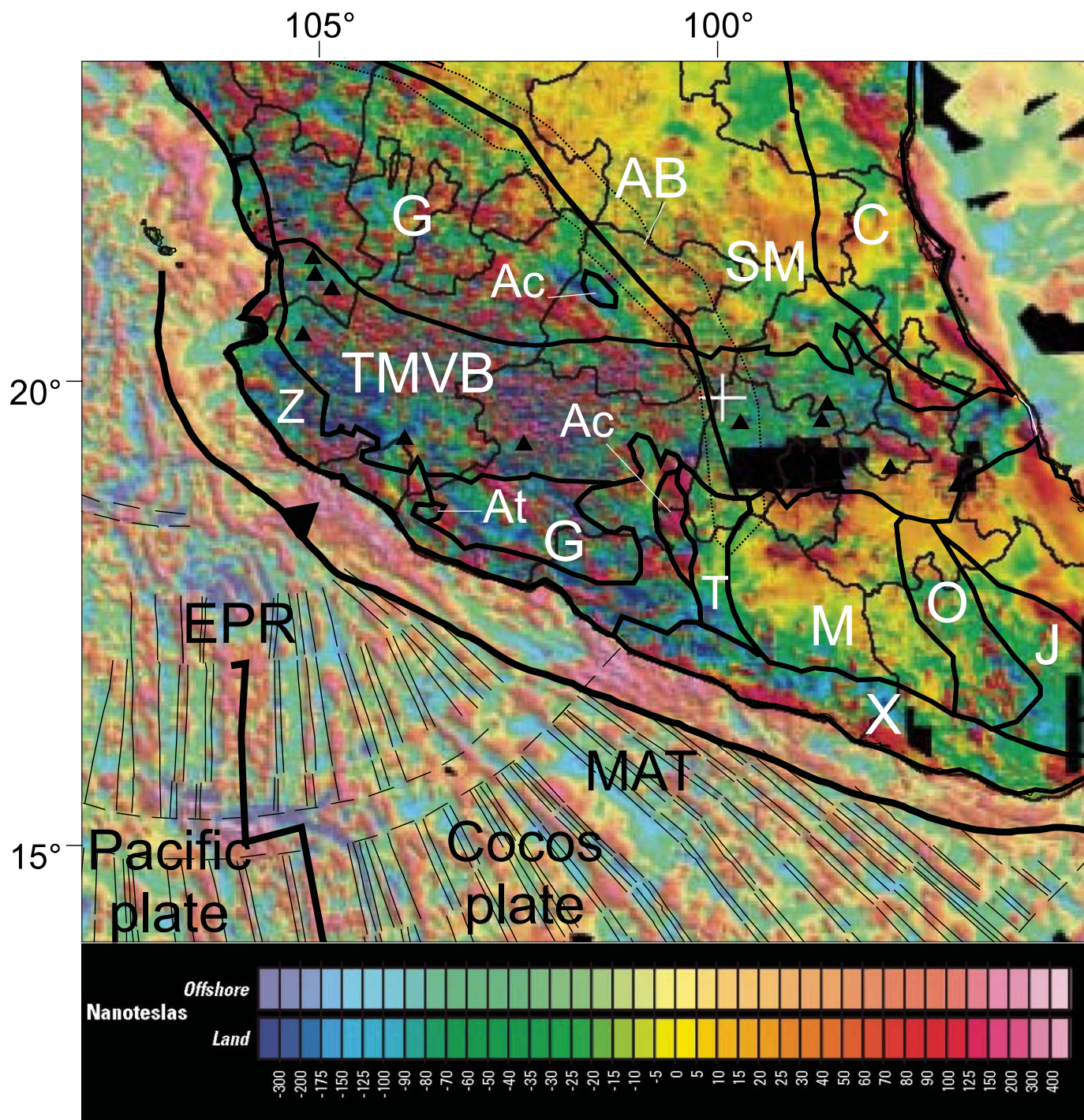


Figure 8. Magnetic intensity map of southern Mexico (data from the North American Magnetic Anomaly Group, 2002). The magnetic map is overlain by terrane boundaries (heavy black lines) from Campa and Coney (1983). Note that an abrupt magnetic contrast marks the suture zone between the allochthonous Guerrero arc terrane to the west with autochthonous, continental terranes of nuclear Mexico to the east. The outline of the Late Cretaceous Arperos basin (AB) formed along the suture zone is shown by the dotted line. Note that the Teloloapan terrane (assumed by some workers to be a Guerrero subterrane) actually lies to the east of the magnetic boundary–suture zone, suggesting that it is distinct from the Guerrero terrane. There is a similar, strong magnetic intensity contrast between the Guerrero arc terrane and nuclear, continental Mexico as that between the arc-related Southern Chortis terrane and the continental Central Chortis terrane (Fig. 3). Mexican terrane abbreviations: C—Coahuila; G—Guerrero; J—Juarez; M—Mixteca; O—Oaxaca; SM—Sierra Madre; X—Xolopa. Guerrero subterrane (Mexico) abbreviations: Ac—Argelia; At—Arteaga; T—Teloloapan; Z—Zihuatanejo. AB—Arperos basin, Mexico; TMVB—Trans-Mexican Volcanic Belt. Thin lines in Mexico are state borders. EPR—East Pacific Rise; MAT—Middle America trench.

TABLE 3. CRUSTAL ELEMENTS OF SOUTHERN MEXICO, CENTRAL AMERICA, AND THE CARIBBEAN (cf. FIGS. 1 AND 6)

Crustal element	Basement age	Names from previous workers
<u>Gondwana elements (continental crust)</u>		
Oaxaca terrane (1)	Grenville	Del Sur (2); Zapoteca (3) Oaxaquia (10)
Mixteca terrane (1)	Paleozoic	Del Sur (2)
Xolopa terrane (1)	Paleozoic	Del Sur (2); Chatino (3)
Central Chortis terrane*	Grenville	Undifferentiated Chortis (2, 3); includes El Tambor
Western Chortis (4); and Motagua (10)		
Northern Chortis metamorphic zone*	Paleozoic	Undifferentiated Chortis (2, 3); northern Chortis, western Honduras borderlands and Explorer province (4)
Tampico block (SE Maya) (2)	Paleozoic	eastern Sierra Madre and northwest Maya (1, 3); Guachichili (3); Chiapas-Peten, Chiapas massif, Yucatan platform, (4)
Teloloapan subterrane (6)	Pre-Mesozoic?	Tierra Caliente complex (7); Nahuatl (3)
<u>Proto-Caribbean elements (transitional to oceanic crust)</u>		
Eastern Chortis terrane*	Jurassic	Undifferentiated Chortis (2, 3); Southern Chortis, northern Mosquitia-Nicaragua Rise basins (4)
Juarez terrane (1)	Jurassic	Cuicateco (3)
<u>Subduction complexes and suture belts</u>		
Arperos basin (2) (closed)	Late Jurassic–Early Cretaceous	Parral, Sombrerete, Toliman (1); western Tepehuano (3); overlaps eastern Guerrero (1, 2) and Nahuatl, Tahue (3)
Motagua (4)	Cretaceous	southernmost Maya (1, 2, 3)
Siuna terrane (5)	Cretaceous	Undifferentiated Chortis (2, 3); Yolaina, southern Mosquitia-Nicaragua Rise basins (4)
<u>Intraoceanic (paleo-Pacific) island arc complex</u>		
Guerrero terrane (1, 2)	Late Jurassic–Early Cretaceous	Nahuatl, Tahue (3)
Arteaga subterrane (6)	Late Jurassic–Early Cretaceous	part of Nahuatl (3)
Zihuatanejo subterrane (6)	Late Jurassic–Early Cretaceous	part of Nahuatl (3)
Arcelia subterrane (6)	Late Jurassic–Early Cretaceous	part of Nahuatl (3)
Southern Chortis terrane*	Mesozoic?	Middle America volcanic province, forearc basin and northern forearc ridge (4); Guerrero (2, 8)
Siuna terrane (5)	Late Jurassic–Early Cretaceous	Undifferentiated Chortis (2, 3); Yolaina, southern Mosquitia-Nicaragua Rise basins (4)
Cuba (amalgamated)	mainly Late Jurassic–Early Cretaceous	Zaza, Pural, Sierra Maestra, Escambray, Las Villas, Cifuentes-Plancetas, Cascarajicara, Organos-Rosario, Isla de Pinos (4)
Cayman Ridge	Tertiary?	Pickle (4)
<u>Oceanic crust</u>		
Cayman Trough	post-Paleocene	
Yucatan Basin	Eocene	
<u>Intraoceanic (paleo-Pacific) plateau</u>		
Caribbean large igneous province	Cretaceous	Columbia basin (4)
Chorotega block (9) Built on CLIP	Cretaceous	Talamanca-Gatun, southern Middle America forearc ridge, San Blas-Darian, (4)
<i>Note:</i> References: (1)—Campa and Coney (1983); (2)—Dickinson and Lawton (2001); (3)—Sedlock et al. (1993); (4)—Case et al. (1984, 1990); (5)—Venable (1994); (6)—Mendoza and Suastegui (2000); (7)—Caballero et al. (2000); (8)—Tardy et al. (1994); (9)—Dengo (1985); (10)—Keppie (2004). Abbreviations: CLIP—Caribbean Large Igneous Province. *Defined in this paper.		

Mexico and the accreted Guerrero arc terrane (Fig. 8) that is similar to the strong contrast between the Southern Chortis terrane of proposed arc affinity and the Central Chortis terrane of known continental affinity (Fig. 3) (Table 2). Restoring the Chortis block to its pre-Tertiary position along the southwest margin of Mexico places what is now the southern Chortis block of Honduras near Acapulco, Mexico, where the Guerrero terrane intersects the Mexican margin (Fig. 7) (Rogers et al., Chapter 5, this volume). This suggests the Southern Chortis terrane is most similar to the Arcelia or Zihuatanejo terranes of the composite Guerrero terrane.

Siuna Terrane—Caribbean Arc

Venable (1994) recognized and defined the Siuna terrane of the Chortis superterrane as an accreted island arc built on oceanic crust and accreted to the southern margin of the Chortis block in the late Cretaceous. Refinements of southern Cordillera and Caribbean evolution by various authors (Tardy et al., 1994; Moores, 1998; Mann, 1999) suggest that the Siuna terrane was the westernmost accreted terrane produced by the collision between the Caribbean arc and the continental margin of Chortis (Rogers et al., Chapter 6, this volume).

Northern Chortis Magmatic Zone—Xolopa Terrane

The Northern Chortis magmatic zone and the Xolopa terrane of Mexico appear to reflect early Tertiary magmatic and metamorphic overprinting (Fig. 1). We assume that the observed overprinting on both terranes suggests that both blocks shared a common history.

In the Northern Chortis magmatic zone the high grade and sheared gneiss with east-west trending left-slip indicators suggests exposure of much deeper crustal levels than observed to the south in the Central Chortis terrane (Southernwood, 1986; Manton, 1987) (Fig. 2). Late Cretaceous and early Tertiary felsic intrusives are common (Fig. 5) in the magmatic zone along with metamorphosed Mesozoic strata including marble containing Cretaceous rudists (Wilson, 1974) (Fig. 2).

In Mexico, the Xolopa terrane displays: (1) progressively increasing metamorphic grade toward the truncated Mexican coast of the Mixteco terrane (Oretega-Gutierrez and Elías-Herrera, 2003); (2) intrusion by plutons of decreasing age to the southeast (Herrmann et al., 1994; Schaaf et al., 1995); and (3) near-vertical southeast-striking mylonitic shear zones with left-slip kinematic indicators (Riller et al., 1992). Keppie (2004) eliminates Xolopa entirely as a terrane. We emphasize the strong similarities remain between this region of Mexico and the Northern Chortis magmatic zone and realign them on the reconstruction shown in Figure 7.

We suggest that both the Northern Chortis magmatic zone and the area of the Xolopa terrane share a genetic origin with the development of a magmatic arc in the early Tertiary oblique to the Farallon-Chortis trench, in a manner similar to the present-day Trans-Mexican Volcanic Belt (Cerca et al., 2004). The obliquity of the early Tertiary arc relative to the trench combined with oblique convergence of Farallon-North America (Engebretson et al., 1985; Schaaf et al., 1995) provides the tectonic setting to translate the Chortis block as a large forearc sliver detached along shear zones developed in the magmatic arc (Riller et al., 1992). This interpretation involving the breaking of a shear zone through the continental arc margin contrasts with the model by Keppie (2004) and Keppie and Morán-Zenteno (2005) for prolonged subduction erosion of a hundreds of kilometers wide swath of the Mexican continental margin.

Tectonic Evolution of the Chortis Block

Restoring the Chortis block to its pre-translation (pre-Eocene) position adjacent to the truncated margin of southwestern Mexico re-aligns the Colon fold-thrust belt of the Eastern Chortis terrane with the fold-thrust belts and ophiolites north of the Motagua suture in Guatemala (Donnelly et al., 1990; Burkart et al., 1994) (Fig. 9A). The present day configuration of these elements is shown in Figure 9B. This reconstruction also provides a best fit of the following elements common to the Chortis block and southwestern Mexico that are shown in Figure 7 and include (1) **influx of late Cretaceous terrigenous sandstone and shale** over early Cretaceous shallow marine platform limestone of both

southern Mexico and Chortis, (2) **Grenville-age basement** common to both areas, (3) **Late Cretaceous shortening** structures common to both areas, and (4) **mid-Cretaceous arc volcanism** that is geochemically similar in both areas.

Three independent lines of evidence support our proposed interpretation shown in Figure 9A that the eastern Chortis block records the collision of the Caribbean arc with the southern margin of North America in the late Cretaceous. The first is the 350 km-long Colon fold belt with Campanian-age, northwest-directed shortening described by Rogers et al. in Chapter 6 of this volume. The second is the spatial association and inferred accretion of the intraoceanic Siuna island arc complex on the southern margin of the Chortis block in the late Cretaceous (Venable, 1994). The third is the Pacific origin of the Caribbean arc and its position at the leading edge of the Caribbean large igneous province (Pindell and Barrett, 1990; Pindell et al., 2006). The entry of this arc-plateau feature into the area of the proto-Caribbean Sea shown in Figure 9A led to the subduction of proto-Caribbean oceanic crust and partial accretion of the Caribbean arc and the oceanic plateau at the “gateways” to the Caribbean in Colombia in South America and in southern Central America where large areas of the crust appear to have been built on oceanic plateau material (Kerr et al., 1998). The Chortis block and southwestern Mexico “corner” shares many common elements to the corner of northwestern South America: in both areas a core of continental rocks is surrounded by collided arc rocks (Fig. 7B and 9A). In the case of Chortis and southwestern Mexico, the arc rocks are not yet as well understood as those in northwestern South America.

CONCLUSIONS

Correlation of a 1985 aeromagnetic survey of Honduras with mapped basement outcrop exposures in Honduras and Nicaragua forms the basis to define three terranes on the Chortis block: (1) **Central Chortis terrane** with exposed Paleozoic basement of continental origin; the northern edge of this terrane is modified by early Tertiary metamorphism and intrusions that overprints areas of the Central and Eastern Chortis terranes; (2) **Eastern Chortis terrane** with a Jurassic metasedimentary basement; and (3) **Southern Chortis terrane** of low magnetic intensity covered by Miocene pyroclastic strata and limited basement exposure. The **Siuna terrane** of Nicaragua of oceanic island origin that accreted to the Chortis block in the late Cretaceous; and the **Northern Chortis metamorphic zone** is defined by early Tertiary metamorphism and intrusions that overprints areas of the Central and Eastern Chortis terranes. These proposed terranes are preliminary but are based on the most up to date compilation of data that is available including the aeromagnetic data. Future mapping and sampling will likely modify the terrane definitions and precisely locate their boundaries with adjacent terranes.

Shared geologic and magnetic characteristics between Chortis and southwestern Mexican terranes—previously described by Campa and Coney (1983), Sedlock et al. (1993) Dickinson

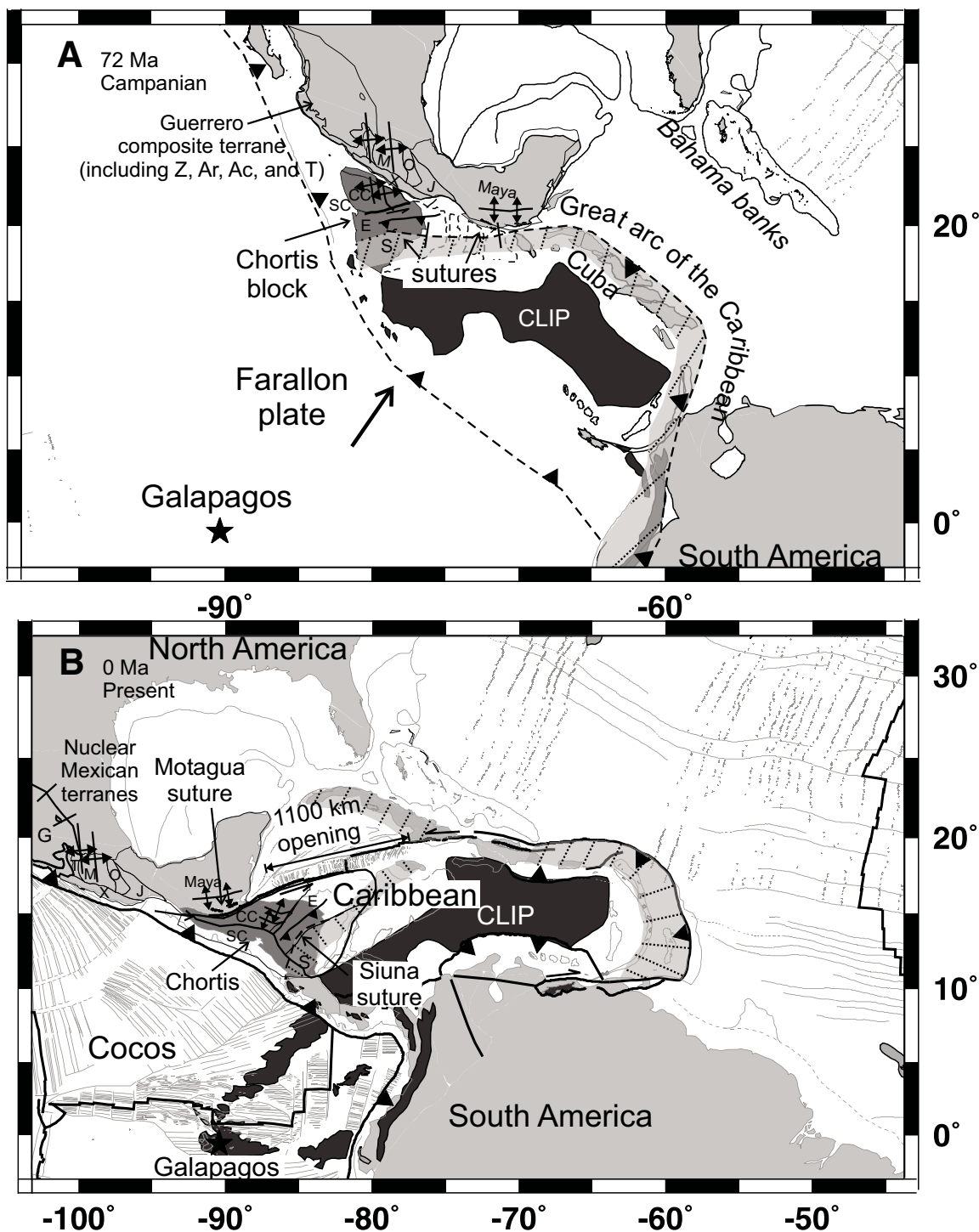


Figure 9. (A) Plate reconstruction of the Chortis block during Campanian time (ca. 72 Ma). This reconstruction uses a mantle reference frame, and the Galapagos hotspot is held fixed. At this time, the Chortis block (medium-gray area) is proposed to form the southwestern continental corner of the North America plate. The Caribbean large igneous province (CLIP; dark area) formed ca. 88 Ma in the eastern Pacific Ocean as part of the Farallon plate and moved northeastward with the Caribbean arc at its leading edge. The Caribbean arc and trailing Caribbean large igneous province collided with the edges of North and South America on either side of the gap (proto-Caribbean Sea) formed by Late Jurassic–Early Cretaceous rifting and oceanic spreading between the two Americas. Accreted parts of the Caribbean large igneous province in South America are shown by the darker areas. Ac—Arcelia; Ar—Arteaga; CC—Central Chortis terrane; E—Eastern Chortis terrane; J—Juarez terrane; M—Maya block; O—Oaxaca terrane; S—Siuna terrane; SC—Southern Chortis terrane; T—Teloloapan; Z—Zihuatanejo. (B) Present-day position of the same tectonic elements shown in A.

and Lawton (2001) and Keppie (2004)—were used to correlate: (1) the Central Chortis terrane to the Mixteca and Oaxaca terranes of Mexico, (2) the Eastern Chortis terrane with the Juarez terrane of Mexico (3) the Southern Chortis terrane to the Guerrero terrane of Mexico, (4) the Northern Chortis metamorphic zone to the Xolopa terrane of Mexico and (5) the Siuna terrane to the circum-Caribbean volcanic arc, or the “Great Arc of the Caribbean” (Pindell and Barrett, 1990; Mann, 1999).

Linking Chortis and Mexican terranes in pre-Tertiary time provides a starting point for improved understanding of the complex geology of terranes in both areas. We present plate reconstructions using all regional plate kinematic and geologic constraints to provide insights into the main tectonic events affecting the Chortis block.

ACKNOWLEDGMENTS

Funding for R. Rogers and P. Mann was provided by the Petroleum Research Fund of the American Chemical Society (grant 33935-AC2 to P. Mann). We thank the Dirección de Energía of Honduras and the Japanese Petroleum Exploration Co. Geoscience Institute for releasing aeromagnetic data used in the study. Bill Dickinson, Ginny Sisson, Randy Marrett, Bill Muehlberger, Ric Finch, Mark Gordon, and Wulf Gose provided many useful discussions on the geology and tectonics of the Chortis block. Special thanks to Lisa Bingham for her help with the text and figures. We thank two anonymous reviews for helpful comments. The authors acknowledge the financial support for publication costs provided by the University of Texas at Austin’s Geology Foundation and the Jackson School of Geosciences. This is University of Texas Institute for Geophysics contribution no. 1869.

REFERENCES CITED

- Arden, D., 1975, Geology of Jamaica and the Nicaraguan Rise, in Nairn, A., and Stehli, F., eds., *The ocean basins and margins*: New York, Plenum Press, v. 3, p. 616–661.
- Avé Lallemant, H., and Gordon, M., 1999, Deformation history of Roatan Island: Implications for the origin of the Tela basin (Honduras), in Mann, P., ed., *Caribbean basins: Sedimentary basins of the world series*: Amsterdam, Elsevier, v. 4, p. 197–218.
- Burkart, B., 1994, Northern Central America, in Donovan, S., and Jackson, T., eds., *Caribbean geology: An introduction*: Jamaica, University of the West Indies Publisher’s Association, p. 265–284.
- Burkart, B., and Self, S., 1985, Extension and rotation of crustal blocks in northern Central America and effect on the volcanic arc: *Geology*, v. 13, p. 22–26, doi: 10.1130/0091-7613(1985)13<22:EAROCB>2.0.CO;2.
- Burkart, B., Denton, B., Dengo, C., and Moreno, G., 1987, Tectonic wedges and offset Laramide structures along the Polochic fault of Guatemala and Chimpas, Mexico: Reaffirmation of large Neogene displacement: *Tectonics*, v. 9, p. 411–422.
- Campa, M., 1985, The Mexican thrust belt, in Howell, D.G., ed., *Tectonostratigraphic terranes of the Circum-Pacific Region*: Houston, Texas, Circum-Pacific Council on Energy and Mineral Resources, Earth Science Series no. 1, p. 299–313.
- Campa, M., and Coney, P., 1983, Tectono-stratigraphic terranes and mineral resource distributions in Mexico: *Canadian Journal of Earth Sciences*, v. 20, p. 1040–1051.
- Cabral-Cano, E., Lang, H., and Harrison, C., 2000, Stratigraphic assessment of the Arcelia-Teloloapan area, southern Mexico: Implications for southern Mexico’s post-Neocomian tectonic evolution: *Journal of South American Earth Sciences*, v. 13, p. 443–457, doi: 10.1016/S0895-9811(00)00035-3.
- Carr, M., Feigenson, M., Patino, L., and Walker, J., 2003, Volcanism and geochemistry in Central America: Progress and problems, in Eiler, J., and Abers, G., eds., *The subduction factory: American Geophysical Union Monograph Series*, v. 138, p. 153–179.
- Case, J., Holcombe, T., and Martin, R., 1984, Map of geological provinces in the Caribbean region: The Geological Society of America Memoir 162, p. 1–30.
- Case, J., MacDonald, W., and Fox, P., 1990, Caribbean crustal provinces: seismic and gravity evidence, in Dengo, G., and Case, J.E., eds., *The Caribbean Region*: Boulder, Colorado, The Geological Society of America, The Geology of North America, v. H, p. 1536.
- Centeno-Garcia, E., Ruiz, J., Coney, P., Patchett, P., and Ortega-Gutierrez, F., 1993, Guerrero Terrane of Mexico: its role in the Southern Cordillera from new geochemical data: *Geology*, v. 21, p. 419–422, doi: 10.1130/0091-7613(1993)021<0419:GTOMIR>2.3.CO;2.
- Cerca, M., Manetti, P., Ferrari, L., Bonini, M., and Corti, G., 2004, The role of crustal heterogeneity in controlling vertical coupling during Laramide shortening and the development of the Caribbean–North America transform boundary in southern Mexico: Insights from analogue models: *Geological Society [London] Special Publication* 227, p. 117–140.
- Cumming, G., Kesler, S., and Krstic, D., 1981, Source of lead in Central America and Caribbean mineralization: *Earth and Planetary Science Letters*, v. 31, p. 262–268.
- DeMets, C., 2001, A new estimate for present-day Cocos–Caribbean plate motion: Implications for slip along the Central American volcanic arc: *Geophysical Research Letters*, v. 28, p. 4043–4046.
- DeMets, C., Jansma, P., Mattioli, G., Dixon, T., Farina, F., Bilham, R., Calais, E., and Mann, P., 2000, GPS geodetic constraints on Caribbean–North American plate motion: *Geophysical Research Letters*, v. 27, p. 437–440, doi: 10.1029/1999GL005436.
- DeMets, C., Mattioli, G., Jansma, P., Rogers, R., Tenorio, C., and Turner, H.L., 2007, this volume, Present motion and deformation of the Caribbean plate: Constraints from new GPS geodetic measurements from Honduras and Nicaragua, in Mann, P., ed., *Geologic and tectonic development of the Caribbean plate boundary in northern Central America: The Geological Society of America Special Paper* 428, doi: 10.1130/2007.2428(02).
- Dengo, G., 1985, Mid America: Tectonic setting for the Pacific margin from southern Mexico to northwestern Columbia, in Nairn, A., and Stehli, F., eds., *The ocean basins and margins*: New York, Plenum Press, v. 7, p. 123–180.
- Dickinson, W., and Lawton, T., 2001, Carboniferous to Cretaceous assembly and fragmentation of Mexico: The Geological Society of America Bulletin, v. 113, p. 1142–1160, doi: 10.1130/0016-7606(2001)113<1142:CTCAAF>2.0.CO;2.
- Dirección General de Minas e Hidrocarburos, 1985, Aeromagnetic survey of Honduras: Tegucigalpa, Honduras, Dirección General de Minas e Hidrocarburos, scale 1:500,000, 1 sheet.
- Donnelly, T., Horne, G., Finch, R., and López-Ramos, E., 1990, Northern Central America: The Maya and Chortis blocks, in Dengo, G., and Case, J., eds., *The Caribbean Region*: Boulder, Colorado, The Geological Society of America, The Geology of North America, v. H, p. 37–76.
- Drobe, J., and Cann, R., 2000, Cu–Au skarn mineralization, Minas de Oro District, Honduras, Central America: *Exploration and Mining Geology*, v. 9, p. 51–63, doi: 10.2113/0090051.
- Emmet, P., 1983, Geology of the Agalteca Quadrangle, Honduras, Central America [M.S. thesis]: Austin, Texas, University of Texas at Austin, 201 p.
- ENEE (Empresa Nacional de Energía Eléctrica), 1987, Geologic investigations of proposed hydrothermal sites, Honduras: Tegucigalpa, Honduras, Honduras Electrical Company, 26 p.
- Engelbreton, D., Cox, A., and Gordon, R., 1985, Relative motions between oceanic and continental plates in the Pacific Basin: The Geological Society of America Special Paper 206, 59 p.
- Fakundiny, R.H., 1970, Geology of the El Rosario Quadrangle, Honduras, Central America [PhD dissertation]: Austin, Texas, University of Texas at Austin, 234 p.
- Finch, R., 1981, Mesozoic stratigraphy of central Honduras: AAPG Bulletin, v. 65, p. 1320–1333.

- Gordon, M.B., 1993, Revised Jurassic and Early Cretaceous (Pre-Yojoa Group) stratigraphy of the Chortis block: Paleogeographic and tectonic implications, *in* Pindell, J.L., and Perkins, R.F., eds., Mesozoic and Early Cenozoic development of the Gulf of Mexico and Caribbean region: A context for hydrocarbon exploration: Austin, Texas, Gulf Coast Section Society of Economic Paleontologists and Mineralogists Foundation, p. 143–154.
- Gordon, M., and Muehlberger, W., 1994, Rotation of the Chortis block causes dextral slip on the Guayape fault: *Tectonics*, v. 13, p. 858–872, doi: 10.1029/94TC00923.
- Gose, W., 1985, Paleomagnetic results from Honduras and their bearing on Caribbean tectonics: *Tectonics*, v. 4, p. 565–585.
- Gose, W., and Swartz, D., 1977, Paleomagnetic results from Cretaceous sediments in Honduras: Tectonic implications: *Geology*, v. 5, p. 505–508, doi: 10.1130/0091-7613(1977)5<505:PRFCSI>2.0.CO;2.
- Guzman-Speziale, M., 2001, Active seismic deformation in the grabens of northern Central America and its relationship to the relative motion of the North America–Caribbean plate boundary: *Tectonophysics*, v. 337, p. 39–51, doi: 10.1016/S0040-1951(01)00110-X.
- Guzman-Speziale, M., and Meneses-Rocha, J.J., 2000, The North America–Caribbean plate boundary west of the Motagua-Polochic fault system, A jog in southeastern Mexico: *Journal of South American Earth Sciences*, v. 13, p. 459–468, doi: 10.1016/S0895-9811(00)00036-5.
- Harwood, R., 1993, Mapa Geológica de Honduras, Hoja de Yuscáran: Tegucigalpa, Honduras, Instituto Geográfico Nacional, scale 1:50,000.
- Herrmann, U., Nelson, B., and Rathschbacher, L., 1994, The origin of a terrane: U/Pb zircon geochronology and tectonic evolution of the Xolapa complex (southern Mexico): *Tectonics*, v. 13, p. 455–474, doi: 10.1029/93TC02465.
- Home, G., Atwood, M., and King, A., 1974, Stratigraphy, sedimentology, and paleoenvironment of Esquias Formation of Honduras: *AAPG Bulletin*, v. 58, p. 176–188.
- Home, G., Clark, G., and Pushkar, P., 1976a, Pre-Cretaceous rocks of northwestern Honduras: Basement terrane in Sierra de Omoa: *AAPG Bulletin*, v. 60, p. 566–583.
- Home, G., Pushkar, P., and Shafiqullah, M., 1976b, Laramide plutons on the landward continuation of the Bonacca ridge, northern Honduras: Guatemala, *Publicaciones Geológicas del ICAITI*, v. 5, p. 84–90.
- Howell, D.G., Jones, D.L., and Schermer, E.R., 1985, Tectonostratigraphic terranes of the Circum-Pacific region: Principles of terrane analysis, *in* Howell, D.G., ed., Tectonostratigraphic terranes of the Circum-Pacific region: Circum-Pacific Council for Energy and Mineral Resources, Earth Science Series, v. 1, p. 3–31.
- INETER (Instituto Nicaragüense de Estudios Territoriales), 1995, Mapa Geológica Minero de la Republica de Nicaragua: Managua, Nicaragua, escala 1:500,000.
- James, K., 2006, Arguments for and against the Pacific origin of the Caribbean Plate: Discussion, finding for an inter-American origin: *Geological Acta*, v. 4, p. 279–302.
- Johnson, C., Lang, H., Cabral-Cano, E., Draper, G., Harrison, C., and Barros, J., 1991, Preliminary assessment of stratigraphy and structure, San Lucas region, Michoacan and Guerrero states, SW Mexico: *Mountain Geologist*, v. 28, p. 121–135.
- Jordan, B.R., Sigurdsson, H., Carey, S., Lundin, S., Rogers, R., Singer, B., and Barquero-Molina, M., 2007, this volume, Petrogenesis of Central American Tertiary ignimbrites and associated Caribbean Sea tephra, *in* Mann, P., ed., Geologic and tectonic development of the Caribbean plate boundary in northern Central America: The Geological Society of America Special Paper 428, doi: 10.1130/2007.2428(07).
- Karig, D.E., Cardwell, R.K., Moore, G.F., and Moore, D.G., 1978, Late Cenozoic subduction and continental margin truncation along the northern Middle America Trench: The Geological Society of America Bulletin, v. 89, p. 265–276, doi: 10.1130/0016-7606(1978)89<265:LCSACM>2.0.CO;2.
- Keppie, J., 2004, Terranes of Mexico revisited: A 1.3 billion year odyssey: *International Geology Review*, v. 46, p. 765–794.
- Keppie, J., and Morán-Zenteno, D., 2005, Tectonic implications of alternative Cenozoic reconstructions for southern Mexico and the Chortis block: *International Geology Review*, v. 47, p. 473–491.
- Kerr, A., Tarney, J., Nivia, A., Marriner, G., and Saunders, A., 1998, The internal structure of oceanic plateaus: Inferences from obducted Cretaceous terranes in western Colombia and the Caribbean: *Tectonophysics*, v. 292, p. 173–188, doi: 10.1016/S0040-1951(98)00067-5.
- Kesler, S., Levy, E., and Martín, F., 1990, Metallogenic evolution of the Caribbean region, *in* Dengo, G., and Case, J.E., eds., The Caribbean Region: Boulder, Colorado, The Geological Society of America, The Geology of North America, v. H, p. 459–482.
- Kozuch, M., 1991, Mapa Geológica de Honduras: Tegucigalpa, Honduras, Instituto Geográfico Nacional, escala 1:500,000.
- Lang, H., Barros, J., Cabral-Cano, E., Draper, G., Harrison, C., Jansma, P., and Johnson, C., 1996, Terrane deletion in northern Guerrero State: *Geofísica Internacional*, v. 35, p. 349–359.
- Leroy, S., Mauffret, A., Patriat, P., and Mercier de Lepinay, B., 2000, An alternative interpretation of the Cayman trough evolution from a reidentification of magnetic anomalies: *Geophysical Journal International*, v. 141, p. 539–557, doi: 10.1046/j.1365-246x.2000.00059.x.
- MacDonald, W.D., 1976, Cretaceous-Tertiary evolution of the Caribbean, *in* Causso, R., ed., Transactions of the 7th Caribbean Geologic Conference, Guadalupe: Paris, Bureau de Recherches Géologiques et Minières Publication 1524, p. 69–78.
- Mann, P., 1999, Caribbean sedimentary basins: Classification and tectonic setting from Jurassic to Present, *in* Mann, P., ed., Caribbean basin, Sedimentary basins of the world series: Amsterdam, Elsevier, v. 4, p. 3–31.
- Manton, W., 1987, Tectonic interpretation of the morphology of Honduras: *Tectonics*, v. 6, p. 633–651.
- Manton, W., 1996, The Grenville of Honduras: The Geological Society of America Abstracts with Programs, v. 28, no. 7, p. A-493.
- Manton, W., and Manton, R., 1984, Geochronology and Late Cretaceous–Tertiary tectonism of Honduras: Tegucigalpa, Honduras, Dirección General de Minas e Hidrocarburos, 55 p.
- Manton, W., and Manton, R., 1999, The southern flank of the Tela basin, Republic of Honduras, *in* Mann, P., ed., Caribbean basins: Sedimentary basins of the world series: Amsterdam, Elsevier, v. 4, p. 219–236.
- Markey, R., 1995, Mapa Geológica de Honduras, Hoja de Moroceli (Geologic Map of Honduras, Morocelli sheet): Tegucigalpa, Honduras, Instituto Geográfico Nacional, escala 1:50,000.
- Marton, G., and Buffler, R., 1994, Jurassic reconstruction of the Gulf of Mexico basin: *International Geology Review*, v. 36, p. 545–586.
- Mendoza, T.O., and Suastegui, M., 2000, Geochemistry and isotopic composition of the Guerrero terrane (western Mexico): Implications for the tectono-magmatic evolution of southwestern North America during the Late Mesozoic: *Journal of South American Earth Sciences*, v. 13, p. 297–324, doi: 10.1016/S0895-9811(00)00026-2.
- Metal Mining Agency of Japan (MMAJ), 1980a, Report on geology survey of the western area (Olancho): Japan International Cooperation Agency, Government of Japan, v. 5, 138 p.
- Metal Mining Agency of Japan (MMAJ), 1980b, Report on geology survey of the western area (consolidated report): Japan International Cooperation Agency, Government of Japan, v. 6, 177 p.
- Mills, R., and Barton, R., 1996, Geology of the Ahuas area in the Mosquitia Basin of Honduras: Preliminary report: *AAPG Bulletin*, v. 58, p. 189–207.
- Mills, R., Hugh, K., Feray, D., and Swolfs, H., 1967, Mesozoic stratigraphy of Honduras: *AAPG Bulletin*, v. 51, p. 1711–1786.
- Moores, E., 1998, Ophiolites, the Sierra Nevada, “Cordillera,” and orogeny along the Pacific and Caribbean margins of North and South America: *International Geology Review*, v. 40, p. 40–54.
- North American Magnetic Anomaly Group, 2002, Magnetic anomaly map of North America: U.S. Geological Survey Special Map, scale 1:10,000,000, http://pubs.usgs.gov/sm/mag_map/.
- Oretega-Gutierrez, F., and Elías-Herrera, M., 2003, Wholesale melting of the southern Mixteco terrane and origin of the Xolapa complex: The Geological Society of America Abstracts with Program, v. 35, no. 4, p. 66.
- Oretega-Gutierrez, F., Ruiz, J., and Centeno-García, E., 1995, Oaxaquia, a Proterozoic microcontinent accreted to North America during the late Paleozoic: *Geology*, v. 23, p. 1127–1130, doi: 10.1130/0091-7613(1995)023<1127:OAPMAT>2.3.CO;2.
- Oretega-Gutierrez, F., Solari, L., Sole, J., Martens, U., Gomez-Tuena, A., Moran-Ical, S., Reyes-Salas, M., and Ortega-Obregon, C., 2004, Polyphase, high temperature eclogite-facies metamorphism in the Chuacús Complex, central Guatemala: Petrology, geochronology, and tectonic implications: *International Geology Review*, v. 46, p. 445–470.

- Pindell, J., and Barrett, S., 1990, Geological evolution of the Caribbean region; a plate-tectonic perspective, *in* Dengo, G., and Case, J.E., eds., *The Caribbean Region: Boulder, Colorado, The Geological Society of America, The Geology of North America*, v. H, p. 405–432.
- Pindell, J., and Dewey, J., 1982, Permo-Triassic reconstruction of western Pangea and the evolution of the Gulf of Mexico/Caribbean region: *Tectonics*, v. 1, p. 179–212.
- Pindell, J., Kennan, L., Stanek, K., Maresch, W., and Draper, G., 2006, Foundations of Gulf of Mexico and Caribbean evolution: Eight controversies resolved: *Geologica Acta*, v. 4, p. 303–341.
- Ranero, C., von Huene, R., Flueh, E., Duarte, M., Baca, D., and McIntosh, K., 2000, A cross section of the convergent Pacific margin of Nicaragua: *Tectonics*, v. 19, p. 335–357, doi: 10.1029/1999TC900045.
- Riller, U., Ratschbacher, L., and Frisch, W., 1992, Left-lateral transtension along the Tierra Colorada deformation zone, northern margin of the Xolapa magmatic arc of southern Mexico: *Journal of South American Earth Sciences*, v. 5, p. 237–249, doi: 10.1016/0895-9811(92)90023-R.
- Rogers, R., 1995, Mapa geológica de Honduras, Hoja de Valle de Jamastran: Tegulcigalpa, Honduras Instituto Geográfico Nacional, escala 1:50,000.
- Rogers, R.D., and Mann, P., 2007, this volume, Transtensional deformation of the western Caribbean–North America plate boundary zone, *in* Mann, P., ed., *Geologic and tectonic development of the Caribbean plate boundary in northern Central America: The Geological Society of America Special Paper 428*, doi: 10.1130/2007.2428(03).
- Rogers, R.D., Mann, P., Emmet, P.A., and Venable, M.E., 2007, this volume, Colon fold belt of Honduras: Evidence for Late Cretaceous collision between the continental Chortis block and intraoceanic Caribbean arc, *in* Mann, P., ed., *Geologic and tectonic development of the Caribbean plate boundary in northern Central America: The Geological Society of America Special Paper 428*, doi: 10.1130/2007.2428(06).
- Rogers, R.D., Mann, P., Scott, R.W., and Patino, L., 2007, this volume, Cretaceous intra-arc rifting, sedimentation, and basin inversion in east-central Honduras, *in* Mann, P., ed., *Geologic and tectonic development of the Caribbean plate boundary in northern Central America: The Geological Society of America Special Paper 428*, doi: 10.1130/2007.2428(05).
- Rosencrantz, E., Ross, M., and Sclater, J., 1988, Age and spreading history of the Cayman trough as determined from depth, heat flow, and magnetic anomalies: *Journal of Geophysical Research*, v. 93, p. 2141–2157.
- Rosenfeld, J.H., 1981, Geology of the Western Sierra de Santa Cruz, Guatemala, Central America: An ophiolite sequence [Ph.D. thesis]: Binghamton, New York, State University of New York at Binghamton, 313 p.
- Ross, M., and Scotese, C., 1988, A hierarchical tectonic model of the Gulf of Mexico and Caribbean region: *Tectonophysics*, v. 155, p. 139–168, doi: 10.1016/0040-1951(88)90263-6.
- Schaaf, P., Morán-Zenteno, D., Hernandez-Bernal, M., Solis-Pichardo, G., Tolson, G., and Kohler, H., 1995, Paleogene continental margin truncation in southwestern Mexico: Geochronological evidence: *Tectonics*, v. 14, p. 1339–1350, doi: 10.1029/95TC01928.
- Scott, R., and Finch, R., 1999, Cretaceous carbonate biostratigraphy and environments in Honduras, *in* Mann, P., ed., *Caribbean basins: Sedimentary basins of the world series: Amsterdam, Elsevier*, v. 4, p. 151–166.
- Sedlock, R., Oretaga-Gutierrez, F., and Speed, R., 1993, Tectonostratigraphic terranes and tectonic evolution of Mexico: *The Geological Society of America Special Paper 278*, 153 p.
- Simonson, B., 1977, Mapa geológica de Honduras, El Porvenir: Tegulcigalpa, Honduras, Instituto Geográfico Nacional, escala 1:50,000.
- Solari, L., Keppie, J., Ortega-Gutierrez, F., Cameron, K., Lopez, R., and Hames, W., 2003, 990 and 1100 Grenvillian tectonothermal events in the northern Oaxaca complex, southern Mexico: Roots of an orogen: *Tectonophysics*, v. 365, p. 257–282, doi: 10.1016/S0040-1951(03)00025-8.
- Southernwood, R., 1986, Late Cretaceous limestone clast conglomerates of Honduras [M.S. thesis]: Dallas, Texas, University of Texas at Dallas, 300 p.
- Sundblad, K., Cumming, G., and Krstic, D., 1991, Lead isotope evidence for the formation of epithermal gold quartz veins in the Chortis block, Nicaragua: *Economic Geology and the Bulletin of the Society of Economic Geologists*, v. 86, p. 944–959.
- Talavera-Mendoza, O., Ruiz, J., Gehrels, G., Meza-Figueroa, D., Vega-Granillo, R., and Campa-Uranga, M., 2005, U-Pb geochronology of the Acatlan Complex and implications for the Paleozoic paleogeography and tectonic evolution of southern Mexico: *Earth and Planetary Science Letters*, v. 235, p. 682–699, doi: 10.1016/j.epsl.2005.04.013.
- Tardy, M., Lapierre, H., Freydis, C., Coulon, C., Gill, J., Mercier de Lepinay, B., and Beck, C., Martinez-R., Talavera-M., Ortiz-H., E., Stein, G., Bourdier, J., and Yta, M., 1994, The Guerrero suspect terrane (western Mexico) and coeval arc terranes (the Greater Antilles and the Western Cordillera of Colombia): A late Mesozoic intra-oceanic arc accreted to cratonic America during the Cretaceous: *Tectonophysics*, v. 230, p. 49–73.
- Venable, M., 1994, A geological, tectonic, and metallogenetic evaluation of the Siuna terrane (Nicaragua) [Ph.D. dissertation]: Tucson, Arizona, University of Arizona, 154 p.
- Viland, J., Henry, B., Calix, R., and Diaz, C., 1996, Late Jurassic deformation in Honduras: Proposals for a revised regional stratigraphy: *Journal of South American Earth Sciences*, v. 9, p. 153–160, doi: 10.1016/0895-9811(96)00002-8.
- von Huene, R., Aubouin, J., Azema, J., Blackinton, G., Carter, J.A., Colbourn, W., Cowan, D.S., Curiale, J.A., Dengo, C.A., Faas, R.W., Harrison, W., Hesse, R., Hussong, D.M., Laad, J.W., Muzylov, N., Shiki, T., Thompson, P.R., and Westberg, J., 1980, Leg 67: The Deep Sea Drilling Project Mid-America Trench transect off Guatemala: *The Geological Society of America Bulletin*, v. 91, p. 421–432, doi: 10.1130/0016-7606(1980)91<421:LTDSDP>2.0.CO;2.
- Weiland, T., Suayah, W., and Finch, R., 1992, Petrologic and tectonic significance of Mesozoic volcanic rocks in the Río Wampú area, eastern Honduras: *Journal of South American Earth Sciences*, v. 6, p. 309–325, doi: 10.1016/0895-9811(92)90049-5.
- Williams, H., and Hatcher, R., 1982, Suspect terranes and accretionary history of the Appalachian orogeny: *Geology*, v. 10, p. 530–536, doi: 10.1130/0091-7613(1982)10<530:STAAHO>2.0.CO;2.
- Williams, H., and McBirney, A., 1969, Volcanic history of Honduras: Berkeley, University of California Publications in Geological Sciences no. 85, 101 p.
- Wilson, H., 1974, Cretaceous sedimentation and orogeny in nuclear Central America: *AAPG Bulletin*, v. 58, p. 1348–1396.

MANUSCRIPT ACCEPTED BY THE SOCIETY 22 DECEMBER 2006